EXPLORING THE PROPERTIES OF COMPLEX THIN FILM AMORPHOUS AND CRYSTALLINE OXIDE DIELECTRICS

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EXPLORING THE PROPERTIES OF COMPLEX THIN FILM AMORPHOUS AND CRYSTALLINE OXIDE DIELECTRICS

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Thin film dielectrics have been and still remain a key component in the semiconductor industrial pipeline. Gaining an understanding of the properties of complex amorphous and crystalline oxides will facilitate their implementation in future technologies. Here, we investigate the effects of dopants on the dielectric and structural properties of various amorphous and crystalline thin films deposited using 90 degree off-axis magnetron cosputtering. We find that doping a small amount of GeO_2 in Ta_2O_5 greatly enhances the dielectric constant, with a maximum dielectric constant of $\varepsilon_r \sim 34$ at a Ta cation fraction of 0.97. In contrast, SiO_2 doping into Ta_2O_5 does not elicit such effect. We determine that the enhancement observed in Ta-Ge-O is due to an enhancement in the ionic polarizability. We follow with a study which probes the atomic structure of amorphous Ta_2O_5 and Ta-Ge-O thin films by systematically applying high energy x-ray diffraction and absorption techniques as well as computational simulations. The diffraction and absorption studies were greatly facilitated in part, by depositing the composition spreads on to amorphous silicon nitride membranes. We then explore the composition dependence of the dielectric properties of $Zr_{1-x}Al_xO_{2-x/2}$ and $Zr_{1-x}Si_xO_2$ to determine the effect of the presence or absence of crystallinity on the dielectric constant and the breakdown field of these materials. Finally, we co-deposit several technologically relevant dielectrics – Hf-Si-O, Hf-Ge-O, Nb-Ge-O, and Ta-Sn-O – to study their dielectric and structural properties as a function of composition and to test whether the type of dielectric behavior observed in the Ta-Ge-O system is a more widespread phenomenon.

BIOGRAPHICAL SKETCH

Taro Albert Naoi was born on February 11, 1986 in Hayward CA to Satoshi and Tomoe Naoi. Taro was a shy and timid little rug rat, but despite his cautious temperament he was occupied by his curiosity even more. As a child he enjoyed releasing his youthful energy at the local park with his friends and letting his imagination run free with his extensive Lego collection. When Taro was five he was joined by his brother, Kenta Thomas Naoi. In contrast to his older brother, Kenta was a gregarious, fearless addition to the family, a suitable partner in crime to the more reserved and cautious older brother.

With age Taro's interests expanded, including piano, soccer, video/computer games, and science fiction. In high school he was, for better or for worse, a carefree student. He enjoyed playing tennis and soccer on the high school team, while sustaining reasonable academic standing, and being involved in the occasional tomfoolery with his good friends. Keeping true to his love for science and technology he pursued a degree in materials engineering in college. There, he made many close friends and formed many memories inside and outside of the classroom.

His initial enthusiasm to attend graduate school was prompted, not by a suggestion from his parents, nor the intention to use it as a resume booster, but by a book by the great 20th century physicist Richard Feynman called, *Surely You're Joking, Mr. Feynman*. This spurred him on to apply and attend an REU and ultimately, matriculate in a graduate program. In the Spring of 2008 he made the decision to leave the sunny beaches of California, heading east to Cornell University with his goal to become an independent thinker.

To family and friends

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CHAPTER 1

INTRODUCTION

The invention of the integrated circuit in the late 1950s can be considered a milestone in modern technology and a start of a new period in modern society: The age of silicon. With the ever-increasing demand for fast, compact, and more efficient electronic devices, there seems to be no slowing down for the development of novel materials such that scaling (reduction of transistor feature size) can continue to support Moores Law. Loosely speaking, Moores Law (or observation) states that the number of transistors on an integrated circuit doubles approximately every two years, though Moores original paper actually described the economics of scaling and the minimum cost per component. [1] Moore's law scaling is typically reserved to describe scaling of computational elements such as metal-organic-semiconductor field-effect transistors (MOSFETs), but memory elements such as dynamic random access memory (DRAM) capacitors experience rapid scaling as well. [2]

Perpetuating Moore's law has largely been possible due to continued materials research and development. This includes the ability to grow pristine single crystalline semiconductors, control the chemistry of the semiconductors through doping, or to grow high quality oxides on the semiconductors. The ability to grow high quality amorphous gate oxides with low defect densities (on the order of $10^{10}/\text{cm}^2$ [3]) is especially important and arguably the major reason silicon has reigned supreme in the semiconductor industry. However, persistent scaling of transistor size introduced a series of problems. One of the "grand challenges" at the turn of the century for the semiconductor industry was finding a suitable replacement for SiO_2. [4]

The critical issue with SiO_2 with transistor scaling is the tunneling of electrons through the gate oxide, which is akin to tunneling of a particle wavefunction through a

finite potential barrier. As we know from quantum mechanics, the probability of the particle wavefunction seeping or "tunneling" through the finite barrier scales exponentially with barrier thickness; a thinner barrier will result in a higher tunneling probability.

The theories which model electron conduction through an oxide have been around for quite a while. For example, calculations performed by Snow and Lenzlinger in the late 60s show that electrons tunneling through a triangular potential barrier follow Fowler-Nordheim tunneling. [5] The problem with this model is that oxides back then were generally much thicker (~1000 Å), in comparison to gate oxides today (sub-100 Å) and at these length scales a full quantum mechanical treatment is necessary. More recently, quantum mechanical modeling of electron conduction through SiO_2 was carried out by Lo et al., and they showed that the gate oxide could scale to 20 Å before tunneling currents became a problem. [6] The ultimate limit to the scaling of SiO_2 before it became incontrovertible problem is still under debate. However, recently Muller et.al. showed using electron energy loss spectroscopy (EELS) that a significant fraction of the gate oxide would consist of interfacial sub-oxide having properties that are different from the bulk reducing the limit to about 1.2 nm for silicon oxide. [7]

Finding a replacement for SiO_2 involved balancing several stringent material requirements, [3, 8] the most critical being the reduction of tunneling currents. Since the tunneling current depends on barrier thickness, an obvious solution is to introduce a thicker gate oxide. We can model the gate oxide as a parallel plate capacitor with gate capacitance, C_{gate}:

$$C_{gate} = \frac{\varepsilon_0 \varepsilon_r A}{t} \tag{1.1}$$

where ε_0 is the permittivity of free space, ε_r the relative permittivity (also known as k or κ, hereupon used interchangeably), A is the area, and t is the thickness of the oxide. Increasing the thickness t requires that one either increase A or ε_r in order to keep the

capacitance constant. Increasing A would defeat the point of transistor scaling, hence increasing ε_r is the obvious choice. Materials with ε_r that are higher than SiO_2 have become known colloquially as "high-k" dielectrics.

It is convenient to define an equivalent oxide thickness t_{eq} (EOT):

$$t_{eq} = \left(\frac{3.9}{k}\right) t_{high-\kappa} \tag{1.2}$$

to facilitate the comparison of oxides of different compositions. Using this parameter one can calculate the thickness of SiO_2 that would achieve an equivalent areal capacitance in comparison to a material with a higher dielectric constant. For instance, if we had a material with a dielectric constant of 39 and $t_{high-\kappa}$ of 40 Å, one could in theory achieve areal capacitances that would be equivalent to using SiO_2 that is 4 Å thick. In reality the dielectric does not scale linearly and have to consider depletion and quantum mechanical effects that can change the effective oxide thickness.

It turns out that it is insufficient to simply increase the dielectric constant of the oxide to reduce tunneling currents. In addition to the dielectric constant, one must also consider the band offsets between the semiconductor and the oxide to prevent Schottky emission of electrons and holes into the oxide bands. [9] This is illustrated in the following relation for the tunneling current expressed in the direct tunneling regime. [10] In simplified form [11] the tunneling current is,

$$J \sim \exp\left(-\frac{8\pi\sqrt{2q}}{3h}(m_{\text{eff}}\,\phi_b)^{0.5}\kappa\, t_{eq}\right) \tag{1.3}$$

where q is the charge, h is Planck's constant, m_{eff} is the effective mass, ϕ_b is the barrier height, κ is the dielectric constant, and t_{eq} is the equivalent thickness. The tunneling current J is a function of ϕ_b and κ so the appropriate figure of merit for the tunneling current is the product $\phi_b^{0.5}\kappa$.

To demonstrate this point, it is useful to plot the dielectric constant versus band gap

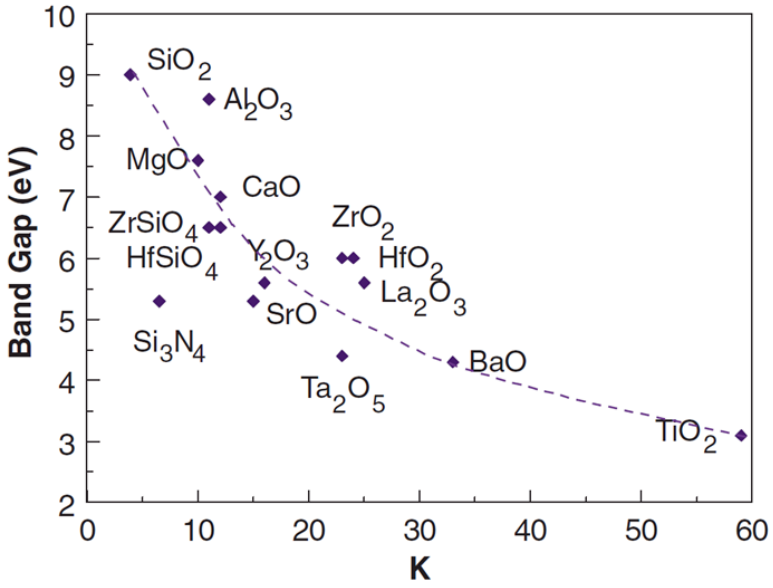


Figure 1.1: Band gap versus dielectric constant for potential high-k gate dielectrics.

(Figure 1.1). [12] Figure 1.1 reveals a roughly inverse dependence on ε_r. Assuming that the band gaps align symmetrically with respect to silicon, this means that oxides with larger dielectric constants will likely have a smaller barrier height and are susceptible to leakage currents by Schottky emission. This suggests that dielectrics such as TiO_2 [13], $BaTiO_3$ [14], and BaSrTiO3 [15, 16] may not be suitable alternatives, even though exhibit very high dielectric constants. In general the conduction band offset is much smaller than the valence band offset, so it is more instructive to plot the dielectric constant as a function of conduction and valence band offsets (Figure 1.2). [12] Ideally, a band offset of at least 1 eV is desirable. In Figure 1.2 we see that La based group III oxides exhibit a good balance of both high dielectric constant and band offset. Indeed, materials such as $LaAlO_3$ [17] and $LaLuO_3$ [18] show promise as potential high-k gate dielectrics.

Other critical factors include interface quality and the thermodynamic and kinetic stability of the oxide in contact with the channel material and the gate electrode. Many potential candidates for gate oxides such as Ta_2O_5 and TiO_2 were initially adopted from

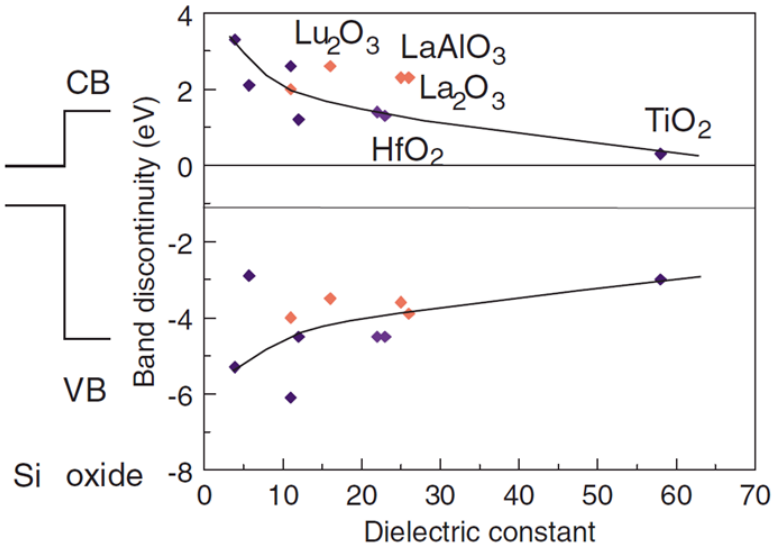


Figure 1.2: Measured band offsets of oxides on Si against dielectric constant for candidate gate oxides. Group III oxides are shown in red, HfO_2 and ZrO_2 in blue. From Reference [12].

DRAM technology, [19, 20] however many turned out to be incompatible as gate oxides due to thermodynamic incompatibility with Si. [21, 22] Achieving a good interface and minimizing the defect densities at the oxide-semiconductor interface is also of paramount concern to abate deleterious effects such as charge trapping which can decrease the carrier mobility in the channel and also cause Fermi level pinning.

1.1 Amorphous transition metal-oxide dielectrics

In the search for a suitable replacement to SiO_2 one consideration is whether to use an amorphous or crystalline material for the gate oxide. The advantage of amorphous materials is the lack of grain boundaries found in polycrystalline materials, which can act as sites for leakage currents to flow. In addition, the applied field will be uniform across the gate because the dielectric constant does not change with crystal orientation of the dielectric. Note, epitaxial growth of single crystalline oxides can also circumvent the problem of grain boundaries and is currently an area of active research. [23]

The challenge with amorphous materials is that many of the potential high-k gate oxides, such as HfO_2 and ZrO_2 crystallize at temperatures required to activate implanted dopants in the channel (~1000 °C). Crystallization of the oxide can be suppressed by doping silicates, but this comes at the cost of a lowered dielectric constant. [24] A surprising result showing an enhanced dielectric constant for zirconium and hafnium silicates at low SiO_2 concentrations, [25, 26] demonstrated the possibility of evading the low-k problem. The result also initiated a slew of experimental and computational attempts to explain the anomalous effect, as it is a peculiar result in amorphous systems. Ultimately, the enhanced dielectric constant were not consistently reproducible, [27, 28] nonetheless it remains to be an interesting theoretical and computational problem. In the following section we provide efforts to understand this anomalous dielectric phenomena in amorphous Hf-Si-O and Zr-Si-O.

1.1.1 Dielectric anomalies in amorphous Hf-Si-O and Zr-Si-O dielectrics

In 2007, Intel announced the introduction of their new transistor incorporating a HfO_2 *based* gate oxide and a metal gate electrode. [29] Work leading up to this involved countless attempts to screen various oxides. From the get-go a prime candidate was transition metal silicates. The rationale behind the silicates was compromising a "high" dielectric constant oxide ($\varepsilon_r \sim 22$ for HfO_2) in exchange for interface quality by doping SiO_2. Initial results for Hf-Si-O and Zr-Si-O were very promising; the oxides did not react in direct contact with Si up to at least 1050 °C, leakage currents were less than $\sim 10^{-6}$ A/cm^{-2} with high breakdown fields on the order of 10 MV/cm. [26] There was evidence that silicates with higher concentrations of transition metal oxide were susceptible to

crystallization at lower temperatures (<1000 °C), [30] but the addition of nitrogen in the metal silicate was shown to improve thermal stability of the material. [31]

The surprising feature though is arguably the unexpected enhancement in the dielectric constant. Referring to the original work by Wilk, [26] a simple calculation reveals that the expected value for ε_r with x ~ 17 at.% Hf in $Hf_xSi_{1-x}O_2$ (again, assuming $\varepsilon_r \sim 22$ for HfO_2 and $\varepsilon_r \sim 3.9$ for SiO_2) is ~ 7. The authors however report a value of ε_r ~ 11, which is about a 60% increase from the expected value based on a weighted mean of the metal oxide concentrations. A similar effect is observed for zirconium silicates, denoting their chemical similarities.

Dielectric enhancement is not unheard of, especially in crystalline form. For example, work by Cava showed that substitution of TiO_2 in Ta_2O_5 stabilizes the H' monoclinic phase resulting in a near four fold increase in the dielectric constant, [32] A similar approach can be taken for Hf and Zr based oxides, where a high temperature phase (tetragonal or cubic) can be stabilized at room temperature to increase the dielectric constant. [33, 34, 35] A perfect demonstration of this is the stabilization of tetragonal HfO_2 by substitution of 5 atomic percent SiO_2 followed by a high temperature anneal. Here, the authors report a maximum κ value of 27, versus an amorphous value of $\kappa \sim 20$. [36]

The relationship between the dielectric constant and structure is more tenuous. This of course is not to say that it is impossible to form a relationship and several experimental and theoretical developments available on this front for hafnium and zirconium silicates. Initial studies utilizing Fourier Transform Infrared Spectroscopy (FTIR) suggested that the enhancement was due to four-fold coordinated Zr-O bonds. [37] In contrast, cluster simulations performed by Kurtz and Devine revealed the energetic stability of six-coordinated Zr over four. [38] A six-coordinated Zr cluster would infer a higher local network density, a smaller molar volume, and thus a higher dielectric constant ac-

cording to the Clausius-Mossotti relation. [38] Calculations by Rigananese et al. related the dielectric constant of Zr-Si-O to local structural units. [27] A modest enhancement of ϵ_r was attributed to ZrO_6 octahedra, but was not sufficient to account for the large reported enhancement in ε_r. Recent large-scale *ab initio* simulations suggest that the inconsistent data for $\varepsilon_r(x)$ in amorphous $Zr_xSi_{1-x}O_2$ and $Hf_xSi_{1-x}O_2$ is due to variability in local structure associated with the variety of coordination numbers accommodated by the transition metal. [28] In that work it was also found that correlation between ε_r and density could not be established, refuting earlier models for dielectric enhancement based solely on changes in density. [38, 39]

The nontrivial dependence of the dielectric constant of Hf(Zr)-Si-O on local structure clearly indicates a wealth of scientific knowledge that can be gained from studying complex amorphous materials. Next, we present a more recent discovery of a tantalum based amorphous oxide exhibiting anomalous dielectric properties that were similar to those observed in the Hf(Zr)-Si-O system. These results leads us to believe that this type of behavior is not entirely unique and that there are other amorphous systems which can exhibit unusual properties.

1.1.2 Tantalum based amorphous materials and anomalous behavior in Ta-Ge-O

Tantalum pentaoxide is a dielectric materials that is used in a variety of technological applications including DRAM memory, antireflective coatings and the more nascent non-volatile memories. [19, 40] Structurally, the material is quite complex, with the existence of many polymorphic structures. This is partly due to Ta's large coordination number which can range from six to as large as eight oxygen atoms. [41, 42] Because

of this complex structural morphology, it is of great interest computationally as well as experimentally. [43, 44, 45, 46]

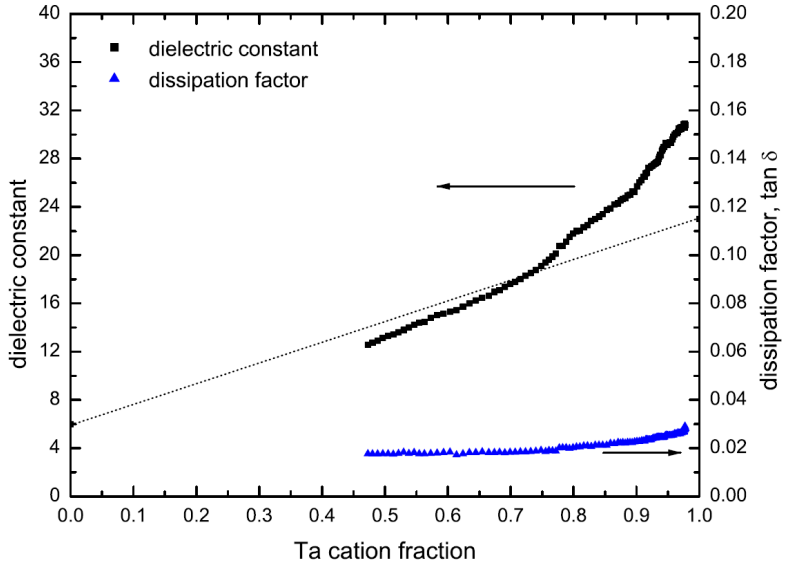


Figure 1.3: The variation in dielectric constant (black squares) and dissipation factor (tan d, blue triangles) of $Ta_xGe_{1-x}O_y$ as a function Ta cation fraction measured at 10 kHz. The dotted line represents the linear interpolation between endmember values. The uncertainty in ε_r is estimated to be ± 5%, dominated by uncertainty in the ellipsometrically determined thickness. From Reference [47].

It was recently reported that mixed amorphous tantalum germanium oxide exhibits a dielectric constant, ε_r as much as 40% higher than that of pure tantalum oxide (Figure 1.3). [47] Moreover, the dielectric constant exceeds the value of both pure Ta_2O_5 and GeO_2. This was not the case for the Hf(Zr) silicates, in which the dielectric constant only exceeded the value of the lower endmember SiO_2. Currently there is no coherent rationale which explains this behavior. Through this dissertation we wish to study this material system further in order to understand the underlying mechanisms which gives rise to this phenomenon.

1.2 Alternative computing and memory architectures: Motivation to study thin film complex amorphous oxides

It is clear that sooner or later Moore's law scaling will end or need to be completely re-defined, either due to reaching a physical limit or it may simply become too impractical to pursue the scaling law. In anticipation of this impending limitation, different avenues are already being explored. These include, but are not limited to advancement in 3D transistors, [48] quantum computing, [49] and neuromorphic computing [50]. Similar trends are being pursued in the memory storage architectures, which emphasize non-volatile forms of memory. [51]

Memory devices expectedly stress different requirements in comparison to logic devices. These are things such as read/write (R/W) times, retention time, number of cycles the memory can endure, energy required to flip the bit, and the maximum achievable memory density. Current R/W times for Hard disk drives (HDDs) are on the order of several milliseconds (Table 1.1). Solid state drives based on Flash(NAND) memory improve R/W times by an order of magnitude or two, but still suffer from low endurance (10^4 cycles). On the other hand, the switching speeds of DRAM capacitors are extremely fast - on the scale of tens of nanoseconds - in addition to exhibiting cycle endurances greater than 10^{16}. The clear drawback however, is the ephemeral nature of the memory (retention time << seconds), since DRAM is not designed for long term memory storage. Comparison of HDD/Flash memory versus DRAM illustrates the historical trend for memory devices to either sacrifice R/W speeds for retention time or retention time for R/W speeds. Emerging memory technologies such as phase change memory (PCM) and spin torque transfer random access memory (STTRAM) attempt to eliminate the need for this trade-off.

	Memristor	PCM	STTRAM	SRAM	DRAM	Flash(NAND)	HDD
		Prototype		Commercialized technologies			
Reciprocal density (F^2)	< 4	4-16	20-60	140	6-12	1-4	2/3
Energy per bit (pJ)	0.1-3	2-25	0.1-2.5	0.0005	0.005	0.00002	$1\text{-}10\times10^9$
Read time (ns)	<10	10-50	10-35	0.1-0.3	10	100,000	$5\text{-}8\times10^6$
Write time (ns)	~10	50-500	10-90	0.1-0.3	10	100,000	$5\text{-}8\times10^6$
Retention time	years	years	years	As long as voltage applied	<< second	years	years
Endurance (cycles)	10^{12}	10^9	10^{15}	$>10^{16}$	$>10^{16}$	10^4	10^4

Table 1.1: Comparison of current memory storage technologies. (PCM, phase change memory; STTRAM, spin torque transfer random access memory; SRAM, static RAM, DRAM, dynamic RAM; HDD, hard disk drive. Taken from Reference [52].

On this front, electrical switches that utilize resistance change to store and perform logic operations (e.g. memristors) are most promising. [52] The concept of switching and memory phenomena involving the differential change in materials resistance has been known for some time. [53] But recent improvements in the ability to probe materials at the nanoscale have facilitated the observation of microscopic changes in the material with switching. One can also draw from the vast knowledge of device processing accumulated over the years to scale devices down to improve memory density. The challenge now, is to gain a deeper understanding of the switching phenomena and relate the structural changes that occur with the electrical properties of the material in order to design suitable materials.

Many resistive materials which display switching behavior start amorphous (mixed amorphous/polycrystalline in the work by Kwon et. al.), [54, 55, 56] which is equivalent to a "0". The material is switched to the "on" state, equivalent to a "1," when a conductive crystalline pathway forms between the electrodes. The exact mechanism for the switching behavior is still under debate, [52] but an understanding of the amorphous structure of amorphous oxides can provide insight into the behavior of such devices. Furthermore, current resistive materials often utilize binary oxides, and thus the resistive

behavior of more complex compositions is not very well understood. To characterize the resistive behavior of complex amorphous oxides, multicomponent sputtering may prove to be useful.

1.3 Deposition technique: off-axis sputtering for multi-composition research

The majority of thin film samples in this work were deposited using RF magnetron sputtering. When considering thin film deposition techniques for creating complex amorphous oxides, sputtering provides several advantages over other deposition techniques which are more commonly found in large scale industrial fabs, namely chemical vapor deposition (CVD) and its close cousin, atomic layer deposition (ALD). CVD methods depend on the thermodynamic reactions between volatile precursors. Therefore, film stoichiometry is strongly predicated on the capacity to control the reaction between precursors using various annealing and gas transport processes. This makes CVD type methods suitable for tightly controlling the stoichiometry of uniform films, but unfit for combinatorial depositions that require depositing many composition in one run.

Sputtering in contrast involves the volatilization of pure metal or ceramic compounds using a plasma, allowing for the deposition of high purity films. When metal targets are used, high purity oxygen is introduced into the chamber to oxidize the volatilized metals. Film stoichiometry can be readily varied across a film in a single run by depositing composition spreads. [57] This is achieved in our lab by orienting the sputtering guns in an oblique configuration, as shown in Figure 1.4.

When the targets are sputtered in this configuration a natural thickness gradient is

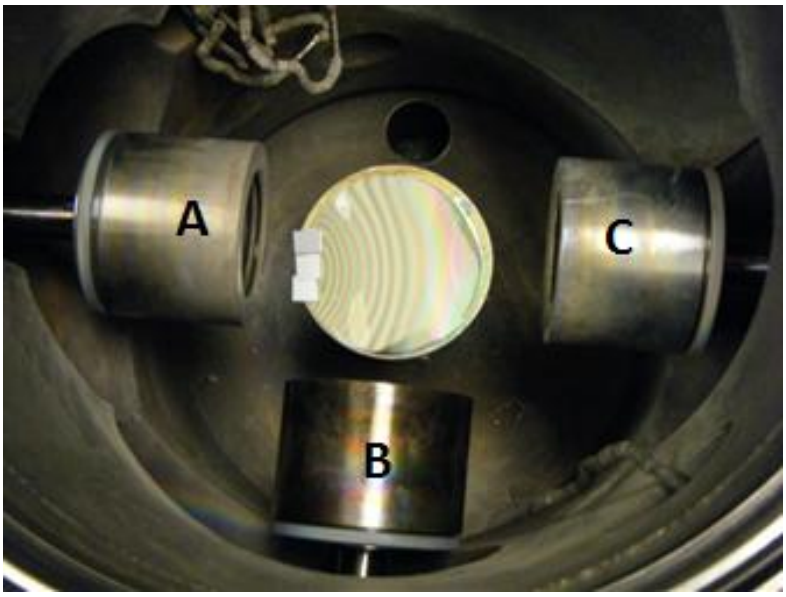


Figure 1.4: Top down view inside the sputtering chamber that is used in this work. Three magnetron sputtering guns are in the 90 degree off-axis configuration with respect to the substrate. Sputtering in this configuration results in a natural thickness gradient across the substrate. Guns A, B, and C can be simultaneously be energized to create a composition spread.

obtained across the substrate. Due to the formation of a natural thickness gradient, sputtering from two or more targets simultaneously will also result in a natural composition gradient. The added bonus here is that that the sputtered constituents are intimately intermixed, i.e. the atoms form a true atomic scale amorphous mixture. To achieve a similar effect using ALD would be cumbersome since it is a technique that involves layer by layer growth. Also in comparison to discrete combinatorial methods [58] the composition resolution of continuous composition methods is much finer, on the order of ~ 0.1 mol%. [57]

The disadvantage of sputtering is that the sputtering can introduce defects and inclusions in the thin film. Not surprisingly the most common inclusion is Ar for sputtering. [59] There is also the effect of introducing substantial film stress in the film if the substrates are biased during the deposition. [60] Nonetheless, applying a substrate bias is a necessary process during off-axis deposition to preclude the formation of columnar

voids. [61] Finally, because sputtering is line of sight deposition technique it is not suitable for depositions that require high conformity to the surface topology of the device. Ultimately, the utility of the materials discovered and evaluated using the composition spread methods need to be compatible with deposition techniques used in industry.

CHAPTER 2

MATERIALS PREPARATION AND ANALYSIS

In this chapter we discuss methods used to prepare and analyze the amorphous oxide films. We first present a theoretical framework for the dielectric response of the materials as well as methods to extract their dielectric properties. Capacitor and silicon nitride membrane fabrication process is then described. Finally, we introduce the techniques used to determine the structure of the non-crystalline films.

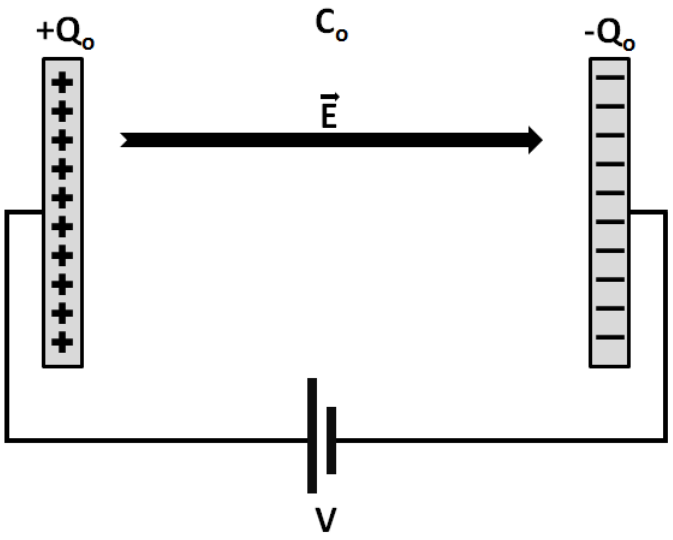
2.1 Insulating materials and charge

Capacitors come in various shapes and sizes, the simplest configuration being the parallel plate capacitor (PPC). The PPC is comprised of an insulating material that is sandwiched between two conducting terminals (Figure 2.1a). If we consider a PPC in a vacuum with only space between the two terminals connected to a battery, the capacitance between the terminals is

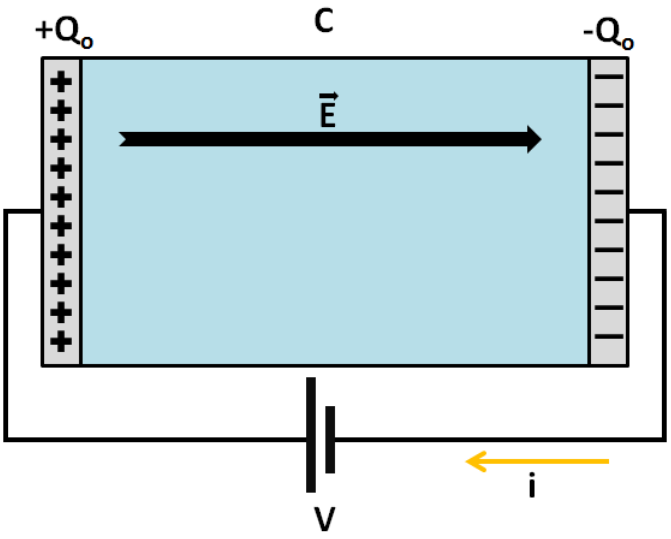
$$C_o = \frac{Q_o}{V}. \tag{2.1}$$

Here, C_o is the capacitance of the parallel plate capacitor in free space, Q_o is the charge on the terminals, and V is the voltage that is applied between the terminals. Now, if we slide an insulating material between the two terminals as shown in Figure 2.1b, current would immediately flow through the external circuit resulting in an increased amount of separated charge. The new charge at the two terminals is now Q such that the capacitance changes to C, so equation 2.1 is slightly modified to

$$C = \frac{Q}{V}. \tag{2.2}$$



(a)

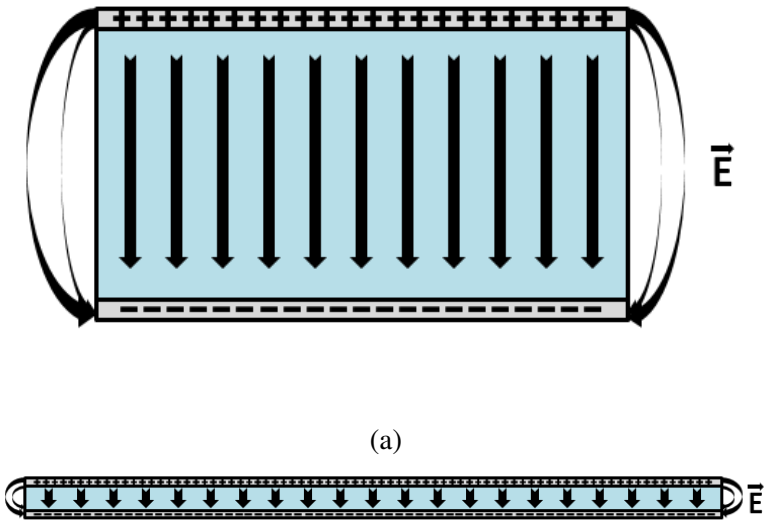


(b)

Figure 2.1: (a) Parallel plate capacitor in a vacuum connected to a battery with voltage V and capacitance C_o (b) The capacitance changes from C_o to C when an insulating material is placed between the terminals.

The ratio between the initial and final charge stored in the capacitor is the relative dielectric constant ε_r (also written as κ or k)

$$\frac{Q}{Q_o} = \varepsilon_r. \tag{2.3}$$



(a)

(b)

Figure 2.2: The contribution of the fringing field to the total electric field changes with the aspect ratio of the capacitor. Thin film capacitors have a large aspect ratio so the contribution of the fringing field is minimal.

and alternatively, the ratio between the two capacitances

$$\frac{C}{C_o} = \varepsilon_r. \tag{2.4}$$

This simple example shows that dielectrics are essentially charge storage devices and that the dielectric constant is a parameterization of how effective a material is able to store charge. Due to its simplicity, many modern electronics exploit the parallel plate configuration. One thing to note is that due to finites size effects the field lines that go from one side of the terminal to the other, especially on the periphery of the terminals, are not perfectly straight lines (Figure 2.2a).

Field lines at the outer edges of the capacitor — so-called fringing fields — are not entirely contained between the plates. For our devices we assume that these fringing fields do not contribute significantly to the electrical measurement owing to the large aspect ratio between the thickness of the insulating film (d) — ranging from ~10 to several 100 nanometers — and the width of the capacitors (w) (several hundred microns)

in our devices, which allows us to greatly simplify the integral equation of Gauss's law. For a capacitor with no dielectric,

$$\oiint_S \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = E_o A = \frac{Q_o}{\varepsilon_o} \tag{2.5}$$

in which E_o is the electric field between the two plates, A is the area of the capacitor, and ε_o is the permittivity of free space. Combining with equation 2.1, rearranging, and from the fact that $E_o = Vd$, we can relate the capacitance to the applied electric field,

$$\frac{\varepsilon_o A E_o}{V} = \frac{\varepsilon_o A}{d} = C_o. \tag{2.6}$$

Recall, that for a capacitor with an insulating material the capacitance is larger by a constant factor ε_r

$$C = \frac{\varepsilon_o \varepsilon_r A}{d} \tag{2.7}$$

Equation 2.7 illustrates the dependence of capacitance on geometry, namely capacitance area and distance between plates, and intrinsic electronic properties through ε_r. Our study focuses on the composition dependence of ε_r for amorphous dielectric oxides. Thus, we introduce the concept of dielectric polarization to describe their intrinsic properties.

2.1.1 Polarization

As we saw earlier, inserting a dielectric between two electrical terminals changes the capacitance between the two terminals. This is a result of the atoms in the dielectric spontaneously polarizing in order to align with the electric field. *Classically*, the spontaneous polarization of the atoms can be represented as a chain of positive and negative point charges that connect head to tail (Figure 2.3). The polarization of the atoms within the material in turn causes the surface of the dielectric to polarize, forming a *surface*

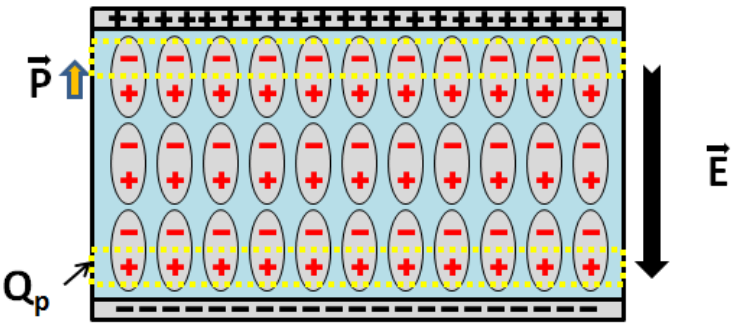


Figure 2.3: Application of an external electric field causes the atoms in the material to polarize along the direction of the applied field. Consequently, a surface polarization charge Q_p forms at the interface between the electrode and the dielectric.

polarization charge Q_p. We can consider the surface polarization charge to be a single dipole moment p across the dielectric. We define polarization to be the total dipole moment per volume, which is equivalently represented as a *surface polarization charge density* σ_p

$$P = \frac{p}{V} = \frac{Q_d d}{Ad} = \frac{Q_p}{A} = \sigma_p \tag{2.8}$$

To compensate for the new charge Q_p that has formed on the surface of the dielectric, current flows through the external circuit (recall Figure 2.1b). As a result, the free surface charge Q_o becomes Q. The total charge on the plates can be expressed as

$$Q = Q_o + Q_p \tag{2.9}$$

If we divide equation 2.9 by the area of the capacitor A, we can rewrite equation 2.9 in terms of charge densities σ

$$\sigma = \sigma_o + \sigma_p \tag{2.10}$$

From equation 2.8 we know that σ_p is the surface polarization density and σ_o is just the *free surface charge density*. The free surface charge density can be written in terms of the electric field, E by rearranging equation 2.5

$$\sigma = \varepsilon_o E + \sigma_p \tag{2.11}$$

Substituting P for σ_p and since $P = \chi_e \varepsilon_o E$ we get

$$\frac{\sigma}{E} = \varepsilon_o(1 + \chi_e) \tag{2.12}$$

where χ_e is the electric susceptibility. Finally, we can relate equation 2.12 back to capacitance

$$\frac{\sigma}{E} = \frac{Q}{AE} = \frac{Cd}{A} = \varepsilon_o(1 + \chi_e) \tag{2.13}$$

$$C = \frac{\varepsilon_o(1 + \chi_e)A}{d} = \frac{\varepsilon_o \varepsilon_r A}{d} \tag{2.14}$$

so $(1 + \chi_e) = \varepsilon_r$.

The simple relation between relative permittivity, capacitance, and susceptibility suggests that measuring the capacitance of the PPC reveals information about the polarization of atoms. For instance, if a polarizable medium did not exist between the conducting terminals (vacuum) $\chi_e = 0$, we would be left with equation 2.6. On the other hand, when an polarizable (e.g. insulator) material was present $\chi_e > 0$ and as such, the capacitance would be greater by a constant ε_r.

2.1.2 Local field, Clausius-Mossotti, and polarizability

We have made the point that macroscopic dielectric properties have microscopic origins. We develop this concept further by establishing a connection between the dielectric constant and polarizability of the atom.

As mentioned earlier, when an electric field $\mathbf{E}_o$ is applied to a dielectric the atoms polarize to align with the field. However, the electric field felt by each atom — the "local field" ($\mathbf{E}_{\text{loc}}$) — is in fact different from the externally applied field, $\mathbf{E}_o$. We calculate the local field by cutting out a hypothetical cavity in the material centered at i, and removing atom i (the atom cannot produce its own field) as shown in Figure 2.4.

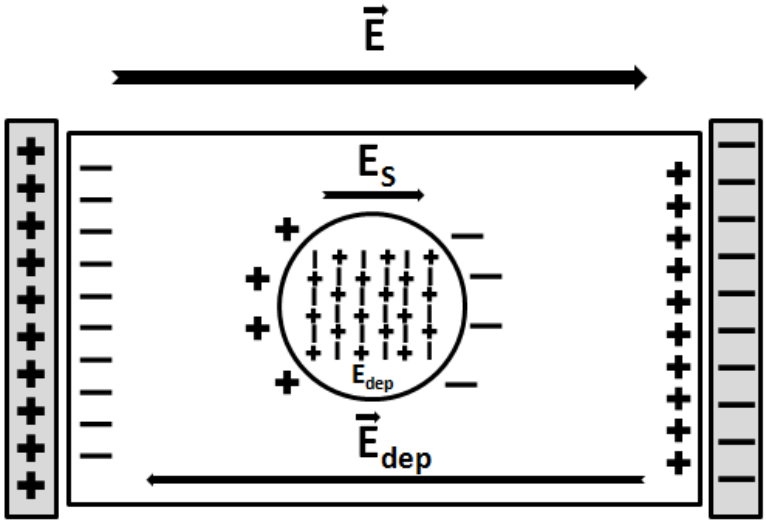


Figure 2.4: To calculate the local field felt by the atom inside the material we create a hypothetical cavity in the material, creating a uniformly polarized sphere. The local field is determined by summing all the contributions to the local field.

The local field at atom i is determined by summing the different contributions to the local field. The contributions from outside of the cavity $\mathbf{E}(\mathbf{r})$ is the difference between the applied field and the depolarization field $\mathbf{E} = \mathbf{E}_o - \mathbf{E}_{dep}$. Within the cavity we have contributions from $\mathbf{E}_{\text{sphere}}(\mathbf{r})$ which is the macroscopic field inside a uniformly polarized sphere, and local dipolar contributions of the atoms within the cavity, $\mathbf{E}_{\text{dipolar}}(\mathbf{r})$. We write $\mathbf{E}_{\text{loc}}(\mathbf{r})$ as

$$\mathbf{E}_{\text{loc}}(\mathbf{r}) = \mathbf{E}(\mathbf{r}) + \mathbf{E}_{\text{sphere}}(\mathbf{r}) + \mathbf{E}_{\text{dipolar}}(\mathbf{r}) \tag{2.15}$$

From electrostatics, we know that $\mathbf{E}_{\text{sphere}}(\mathbf{r})$ is $4\pi\mathbf{P}(\mathbf{r})/3$, where $\mathbf{P}(\mathbf{r})$ is the polarization density defined as $\mathbf{P}(\mathbf{r}) = \mathbf{p}(\mathbf{r})/v$, that is, polarization per unit volume. Equation 2.15 is further simplified for high symmetry materials. For these materials, e.g. cubic and amorphous materials, we assume that dipolar term, $\mathbf{E}_{\text{dipolar}}(\mathbf{r})$ vanishes. Now, the local field is

$$\mathbf{E}_{\text{loc}}(\mathbf{r}) = \mathbf{E}(\mathbf{r}) + \frac{4\pi\mathbf{P}(\mathbf{r})}{3} \tag{2.16}$$

The following,

$$\mathbf{D}(\mathbf{r}) = \varepsilon \mathbf{E}(\mathbf{r}) \tag{2.17}$$

relates the electric displacement $\mathbf{D}(\mathbf{r})$ to the dielectric constant ε. Using Equations 2.16 and 2.17, we can express $\mathbf{P}(\mathbf{r})$ in terms of $\mathbf{E}(\mathbf{r})$.

$$\mathbf{P}(\mathbf{r}) = \frac{\varepsilon - 1}{4\pi}\mathbf{E}(\mathbf{r}) \tag{2.18}$$

Combining Equation 2.16 and 2.18 eliminates $\mathbf{P}(\mathbf{r})$ and becomes an equation that links the local electric field $\mathbf{E}_{\text{loc}}(\mathbf{r})$ to measurable quantities.

$$\mathbf{E}^{\text{loc}}(\mathbf{r}) = \frac{\varepsilon + 2}{3}\mathbf{E}(\mathbf{r}) \tag{2.19}$$

Polarization of the atom is related to the polarizability and the local field and polarizability by $\mathbf{p}(\mathbf{r}) = \alpha \mathbf{E}_{\text{loc}}(\mathbf{r})$ and recalling that $\mathbf{P}(\mathbf{r}) = \mathbf{p}(\mathbf{r})/v$, we can use Equation 2.18 and 2.19 to obtain the Claussius-Mossotti equation.

$$\frac{\varepsilon - 1}{\varepsilon + 2} = \frac{4\pi\alpha}{3v} \tag{2.20}$$

The Claussius-Mossotti (CM) equation is a much used equation that relates a local property — polarizability, α — to a macroscopic property, the dielectric constant.

In actuality, the situation is a bit more subtle. A more thorough investigation of the phenomena of polarization reveals the ambiguity of assigning individual dipole moments to atoms and the choice of a proper primitive cell, known as the polarization ambiguity. There have recently been several theoretical developments that have led to sophisticated methods allowing researchers to calculate the spontaneous polarization of periodic solids. [62] However, the aperiodicity of amorphous materials still poses a theoretical and computational challenge due to the lack of long range ordering. In this study, we merely rely on the CM equation as a way to experimentally extract trends in how the polarizability (a local property) changes as a function of concentration.

2.1.3 Frequency dependence

The process of polarization is an inherently frequency dependent process. Different polarizability mechanisms are accessible at different frequencies and gives rise to the dispersive behavior of the dielectric constant (Figure 2.5).

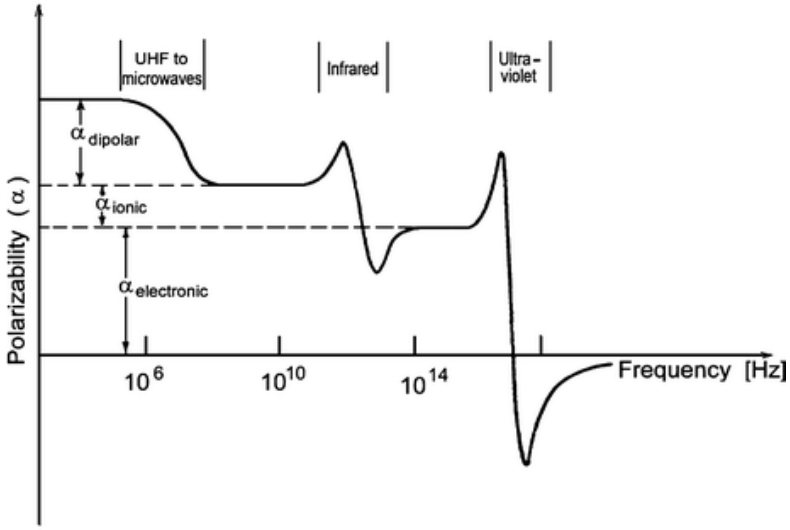


Figure 2.5: Dispersion relation of polarizability as a function of frequency. Figure taken from reference [63].

For dielectric materials that are presented in this study, the components that contribute to the dielectric constant are mainly electronic ($\alpha_{electronic}$) and ionic (α_{ionic}) polarizabilities. The total polarizability can be described as a linear combination of these mechanisms

$$\alpha_{total} = \alpha_{electronic} + \alpha_{ionic} \tag{2.21}$$

and are both roughly active up to infrared frequencies ($\sim 10^{12}$ Hz) (Figure 2.5).

Accurate modeling of polarizabilities requires detailed theoretical considerations, but we can describe to first order their behavior using crude models. Using these crude models the electronic and ionic polarizabilities are written as

$$\alpha_{electronic} = \frac{Z_i e^2}{m(\omega_o^2 - \omega^2)} \qquad \alpha_{ionic} = \frac{e^2}{M(\bar{\omega}^2 - \omega^2)} \tag{2.22}$$

where Z is the atomic number, e the electric charge, $m(M)$ is the mass of the electron(ions), $\bar{\omega}$ is the resonance frequency of the crystal, ω_o is the frequency of the electronic shell, and ω is the driving frequency of oscillation. It is not surprising that these two equations are similar in form since they are both theoretically based on frequency dependent displacement of mass (similar equations of motion). Another point is that the resonance frequency $\bar{\omega}$ of the lattice is considerably smaller in comparison to the frequency of the electronic shell, ω_o. This gives rise to the dispersive behavior of α because the rate at which the denominator increases for α_{ionic} is much slower compared to $\alpha_{electronic}$.

According to Equation 2.20, an anomalous change in the dielectric constant would require that either the electronic or ionic term of the polarizability be greatly enhanced, or a substantial in the molecular volume. It is more likely however, that large enhancements are a result of anomalous behavior of the phonons, which is related to α_{ionic}. For special classes of materials such as ferroelectrics, the softening of certain modes is an indication of a ferroelectric phase transition. [64] We refer to the Lydanne-Sachs-Teller relation,

$$\frac{\omega_{LO}^2}{\omega_{TO}^2} = \frac{\varepsilon(\omega = 0)}{\varepsilon(\omega = \infty)} \tag{2.23}$$

to show that at the ferroelectric transition point, the static dielectric constant, $\varepsilon(\omega = 0)$ diverges, implying that $\omega_{TO}^2 \to 0$. The softening of phonon modes can also be observed in non-ferroelectric materials such as quartz. The point here is that since the ionic polarizability of the material is intimately related to the phonons, an enhancement in the dielectric constant would indicate an enhancement in α_{ionic}.

At frequencies beyond $\sim 10^{12}$ Hz the nucleus of the atom is too sluggish to respond

to the alternating electric field thus, ionic polarization becomes negligible. The total polarization reduces to just the electronic component and the Clausius-Mossotti relation reduces to the Lorentz-Lorenz equation.

$$\frac{n^2 - 1}{n^2 + 2} = \frac{4\pi\alpha_{electronic}}{3v} \tag{2.24}$$

We can obtain the refractive index using optical techniques and x-rays to determine the molecular volume of the material.

To capture the inherent frequency dependent nature of the dielectric constant in experiment, we write ε_r as a complex function,

$$\varepsilon_r = \varepsilon_r' - i\varepsilon_r'' \tag{2.25}$$

where ε_r' and ε_r'' is the real and imaginary part of the dielectric constant respectively. The real part of the dielectric constant is ε_r' is the measured relative permittivity in for example, the capacitance in a PPC. On the other hand, ε_r'' represents the energy lost during the polarization process for instance by collision with neighboring atoms resulting in energy lost as heat.

We define the loss tangent, $\tan\delta$ (also known as the dissipation factor), which equals the ratio between the real and imaginary parts of the dielectric constant.

$$\tan\delta = \frac{\varepsilon_r''}{\varepsilon_r'} \tag{2.26}$$

A high dissipation factor implies that the lossy component of the dielectric constant is large. The lossiness of the dielectric is usually a symptom of poor film quality, i.e. the presence of defect modes and are generally undesirable because they exhibit a significant amount of dispersion in the dielectric constant. Modern devices require high quality control and stability of electronic properties over a large frequency range, so we would like to minimize the frequency dispersion of the oxide. We therefore require films to

have a low dissipation factor (<0.05). In the following section, we show how the dielectric constant and the dissipation factor are determined from electrical measurements.

2.2 Characterization of dielectric properties

Capacitance measurements were performed using a semi-automatic probe station controlled by Labview software, capable of mapping the capacitance and dissipation of an entire 3 inch wafer. Contact to the Pt top electrode was made using a BeCu probe tip, while gold wire was soldered to an exposed portion of the Pt bottom electrode with indium. The probe and gold wire are both connected to a Hewlett Packard 4284A inductance-capacitance-resistance meter (LCR) with a stripped coaxial cable. The LCR meter works by applying AC voltage (V) and recording the current (I) which is used to calculate the capacitance and dissipation factor based on their phase relation. In this study, the applied AC voltage was on the order of 100 mV.

Both current and voltage are complex quantities and thus can be expressed as complex exponentials with phase lag θ between the two quantities,

$$V = |V|e^{j\omega t} \tag{2.27}$$

$$I = |I|e^{j(\omega t+\theta)} \tag{2.28}$$

In an ideal capacitor the current leads the voltage by $\pi/2$ so any deviation from this value is symptomatic of a non-ideal (real) capacitor. As discussed earlier, the impedance of a real capacitor has contributions from lossy and non-lossy components. Hence, the ratio between the current and voltage is the complex impedance and can be expressed in the following way:

$$\frac{V}{I} = Z = R + jX_c \tag{2.29}$$

Here, R represents the real portion of the impedance (resistance), X_c is the imaginary component known as the reactance, and j is the imaginary number $j = \sqrt{-1}$. We can represent the quantity Z as a vector composed of real and imaginary components R and X_c as in Figure 2.6.

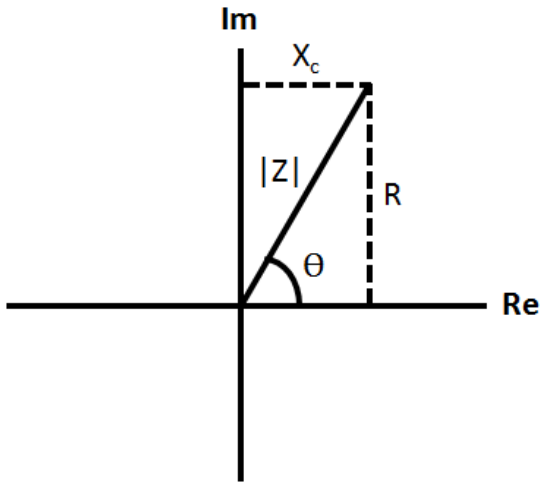


Figure 2.6: Complex impedance.

The magnitude of the impedance $|Z|$ and angle θ denoting the phase difference between R and X_c is,

$$|Z| = \sqrt{R^2 + X_c^2} \tag{2.30}$$

$$\theta = \tan\left(\frac{X_c}{R}\right)^{-1} \tag{2.31}$$

For sake of mathematical simplicity, it is convenient to write the impedance as admittance Y, which is just the inverse of Z,

$$\frac{1}{Z} = Y = G + jB_c \tag{2.32}$$

such that now $|Z|$ and θ are

$$|Z| = \sqrt{G^2 + B_c^2} \tag{2.33}$$

$$\theta = \tan\left(\frac{G}{B_c}\right)^{-1} \tag{2.34}$$

in which G is known as the conductance and B_c is the susceptance with value $|\omega C|$. Finally, the dissipation factor D is,

$$D = \tan\delta = \tan\frac{|\omega C|}{G} \tag{2.35}$$

and is related to θ by $\delta = \pi/2 - \theta$.

2.2.1 Optical measurements

At optical frequencies the atomic nuclei are too sluggish to respond to the alternating electric field, thus only the electronic response contribute significantly to the relative permittivity. The dielectric response at optical frequencies is expressed as the complex refractive index $\tilde{n} = n+ik$, which is a complex quantity written in terms of the absorption coefficient k, and n, the ratio between the speed of light in a vacuum and the speed of light in a medium. We can determine n and k as well as geometric properties such as film thickness by conducting optical measurements; in our case it is variable angle spectroscopic ellipsometry (VASE).

Ellipsometry works by shining light on to the surface of the material and measuring the change in polarization of the light after it has interacted with the material. For a single interface, we can quantify the polarization of the light in terms of the Fresnel reflection coefficients (r_p) and (r_s), the p and s-polarization components respectively.

$$r_{12}^{p} = \frac{\tilde{N}_2\cos\phi_1 - \tilde{N}_1\cos\phi_2}{\tilde{N}_2\cos\phi_1 + \tilde{N}_1\cos\phi_2} \qquad r_{12}^{s} = \frac{\tilde{N}_1\cos\phi_1 - \tilde{N}_2\cos\phi_2}{\tilde{N}_1\cos\phi_1 + \tilde{N}_2\cos\phi_2} \tag{2.36}$$

In the equation, $\tilde{N}_1$ and $\tilde{N}_2$ denote the complex refractive index of materials 1 and 2, and ϕ_1 and ϕ_2 are the angle of incidence and transmission from medium 1 to medium 2 respectively. Now, to describe a two interface material the total reflection coefficients

for the p and s-polarized components R_p and R_s can be explicitly written as,

$$R^p = \frac{r_{12}^p + r_{23}^p e^{-j2\beta}}{1 + r_{12}^p r_{23}^p e^{-j2\beta}} \qquad R^s = \frac{r_{12}^s + r_{23}^s e^{-j2\beta}}{1 + r_{12}^p r_{23}^s e^{-j2\beta}} \tag{2.37}$$

The subscripts denote the incident medium and transmitted medium, while $\beta = 2\pi\frac{d}{\lambda}\tilde{N}_2 \cos\phi_2$ and describes the phase shift of the light that occurs when passing through the various mediums.

We can combine these quantities in the fundamental equation of ellipsometry.

$$\frac{R_p}{R_s} = \tan(\psi)e^{i\Delta} \tag{2.38}$$

The amplitude ratio of reflection, $\tan(\psi)$, and phase shift, Δ, can be measured experimentally, however relevant optical and geometric parameters such as film thickness and refractive index are calculated by fitting to a model. Note, the variability of the angle and the spectroscopic nature of the measurement increases the number of fitted parameters, making the fitting process more robust. For many optically transparent non-absorbing materials — oxides being one of them — the Cauchy relation provides a simple empirical solution. The Cauchy dispersion relation is,

$$n = A + \frac{B}{\lambda^2} + \frac{C}{\lambda^4} \tag{2.39}$$

The capital letters are fitted parameters and λ is the wavelength of light. For most materials a two term model (one utilizing only parameters A and B) is enough to fully describe the dispersion relation of the material. The recursive fitting procedure is performed in the VASE software package provided by Woollam Inc.

2.3 Synthesis of materials and devices

Metal-insulator-metal (MIM) capacitors were fabricated and evaluated combinatorially to explore new amorphous dielectrics as well as characterize known systems. The ad-

vantage of the MIM structure is its simplicity and the flexibility to assess a wide range of technologically relevant metrics. Our studies were mainly concerned with the dielectric properties of amorphous oxides and as such, we used the dielectric constant as a figure of merit when evaluating the MIM capacitors. To determine the resistive behavior of resistive memory devices, the MIM can also potentially be adapted to determine the resistive behavior of the material by measuring the I-V curve.

The majority of MIM capacitors were fabricated on 75 mm, 350 μm thick (100) Si wafers utilizing two deposition methods: electron beam (e-beam) evaporation for the deposition of metal electrodes and rf magnetron sputtering to deposit the oxide films. The general procedure is as follows. Blank Si wafers were first coated with 300 Å Cr adhesion layer using e-beam evaporation. Then without breaking vacuum, a 300 Å Pt base electrode layer was deposited, also using e-beam evaporation. The samples were removed from the evaporator and transferred to the sputtering chamber. Whole wafers were typically used, but in some cases cleaved to smaller pieces in order to accommodate for the size limitation of some measurement apparatuses. The chamber is capable of simultaneously depositing from up to three sputtering targets in the "off-axis" configuration as shown in Figure 2.7.

Here, off-axis means the target face is oriented perpendicular to the sample surface. Deposition from a single target in this configuration results in a natural thickness gradient across the wafer due to the exponential decrease in deposition rate with distance. Thus, when multiple sputtering guns are active a natural composition gradient is obtained. Because of the intimate intermixing of the sputtered constituents, the compositional resolution that can be achieved is typically better than < 0.1 mole-%. Note, the chamber is capable of sputtering in the conventional "on-axis" configuration as well, where the target face is oriented parallel to the sample surface. This orientation was

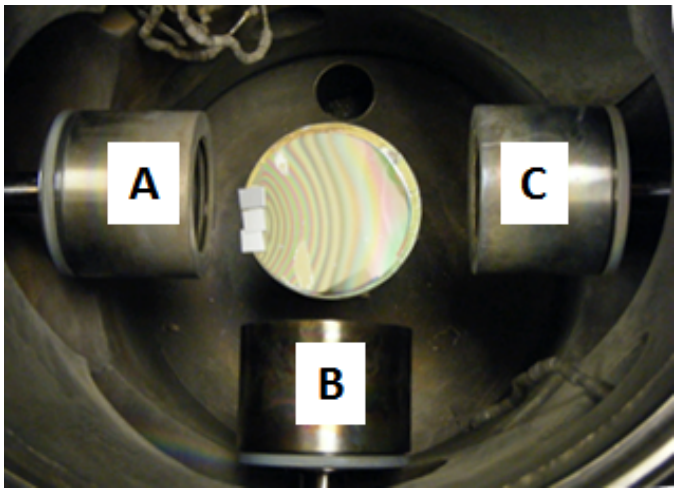


Figure 2.7: Off-axis configuration of the guns enable combinatorial studies of thin film amorphous oxides. Up to three sputtering guns can be simultaneously energized to form a ternary composition. RF biases ares applied to the substrate to form a dense film.

preferred when single composition films of uniform thickness was required.

Base pressures down to 10^{-6} Torr were achieved by using a turbo molecular pump that is backed by a roughing pump. A gate valve is throttled to control the chamber pressure during deposition. Film thicknesses generally varied from approximately 10 to 300 nm. An rf bias was often applied to the films during off-axis deposition to preclude the formation of voidy films. These voids have been show to exhibit defect mediated polarization mechanisms that are undesirable for material integration into practical devices. [61] Once the films were deposited the samples were transferred back to the evaporation chamber. This time, 300 Å Pt dots were deposited on to the sample by evaporation through a shadow mask with 200 μm diameter holes arranged in a square array with center to center spacing of 500 microns.

2.3.1 Application of e-beam evaporation of metal electrodes and electrode material choices

E-beam evaporation is a technique that uses high energy electron beams to evaporate source materials in a vacuum. Typically the evaporators are encased in a bell jar, and evacuated to $\sim 10^{-7}$ Torr prior to deposition. A large current is passed through a metal filament causing electrons to boil off thermionically and accelerate towards the source material, which is resting inside a water cooled hearth. The electrons are steered by the magnetic field produced by a solenoid. The force on the electron can be calculated using the Lorentz force equation, $\mathbf{F} = q_e\mathbf{E} + q_e\mathbf{v} \times \mathbf{B}$. The field can also be modulated so that the electron beam rasters across the source material, improving deposition uniformity. The main advantage over thermal evaporation is that e-beam evaporation can be used to evaporate most any material, even highly refractory metals. Although the energy input of the electron beam on the source material far exceeds the strength of the chemical bonds of the material, the overall process is still thermal. Thus the atoms are deposited with energy of ~ 0.2 eV. The fact that it is a line of sight process is also favorable for the use of a shadow mask and minimizing the formation of penumbra caused by insufficient shadowing.

Some consideration is required when choosing the material that is to be used as electrodes. Because we are measuring the electrical properties of thin film amorphous oxides using MIM structures, we would like to minimize the amount of interfacial oxide formation that would result in inaccurate characterization of the thin film. As an electrode material, platinum is an obvious choice because it is a chemically inert material. However, silicon has a native surface oxide that inhibits the direct deposition of noble metals such as Au and Pt directly on to Si because of their different bonding characteristics. In such a case, a metal that is chemically more reactive needs to be deposited

beforehand as an adhesion layer. In this study 300 Å layer of Cr was commonly used, but other "reactive" metals such as Ti would also suffice. A thickness of 300 Å is more than enough material to yield a continuous layer.

2.3.2 Reactive radio frequency magnetron sputtering and the cosputtering technique

Sputtering, like e-beam evaporation is an energy vaporization method, making it suitable for depositing a wide range of materials. However, evaporation is a thermal process where as sputtering involves momentum transfer of partially ionized gas particles in the form of a plasma. Conservation of energy and momentum tells us that if two particles are of similar mass, then only a small amount of energy is required to displace the particles from their rest state. For this reason, inert gases such as Ar, Kr, and Xe are suitable choices as impinging particles because of their comparable masses to the target elements. Furthermore, they are generally inexpensive and high in purity. For this study, we used Ar — specifically Ar^+ — as the impinging gas.

Energy input of ionized gas ions in sputtering usually range from 10 to a few 100 eV. Low energy ions can be used for surface modification or densification of films, [65, 66] while higher energies are required for sputtering. Dislodged (sputtered) target atoms typically acquire kinetic energies up to several electron volts and undergo various reactions such as colliding with, hitting reactive gasses (reactive sputtering) generating secondary electrons before eventually condensing on to the substrate. The final structure and morphology of the thin film ultimately depends on several tunable parameters.

Structural control can be achieved using various flavors of sputtering. In our study

we mainly use rf magnetrons. Rf sputtering is a flexible technique that can be used to deposit from both metals and insulating targets. Sinusoidal power is applied to the driven electrode, with frequency of oscillation typically at 13.56 MHz. Initially during the positive half of the cycle, the electrons respond to the electric field and are drawn to the cathode. However, when the electrode is driven negative, the ions are unable to respond to the sudden change in electric field due to their large size. Thus, the amount of ion current that flows through the target is significantly smaller in comparison to the electron current. Flow of excess electron current is most often impeded by the use of a blocking capacitor. As time progresses the power and the coupling capacitor continues to charge negatively (almost instantaneously). Eventually a steady state is reached and the target is negatively self-biased. Maximum power delivery to the target is accomplished by means of matching network that works in parallel with the blocking capacitor.

Use of planar magnetron sputtering guns instead of planar-diode further improves sputtering efficiency. Powerful circular magnets embedded in planar magnetron guns produce magnetic field lines that emanate out into the chamber and curl around back into magnets at the center of the gun. When an electric field is simultaneously applied across the planar cathode, it creates a region where the electric and magnetic field cross perpendicularly. Electrons in this region travel in a cycloidal pattern and become confined to that region . Consequently, ionization of gas atoms is most efficient there, producing a circular race-track erosion pattern. Magnetron sputtering has the effect of nonuniform erosion patterns across the target surface, but the benefits are that a low working pressure that is required to sputter materials. Therefore the deposition rate is increased due to decreased collisions and a lower voltage is required to sustain the plasma due to the confinement of electrons.

Most films in this study were sputtered reactively; films were deposited from metal

targets in the presence of a working and reactive gas. The gaseous cocktail consisted of Ar as our working gas and O_2 as the reactive component. Reactive sputtering has several advantages over sputtering directly from a compound (ceramic) target. For one, metal targets can be sputtered at a higher deposition rate compared to ceramic targets and are not as brittle and prone to thermal shock. Higher target purity is also achievable with metal targets so in theory the deposited films should be of higher purity. However, because the target atoms must undergo reactions with the reactive gas to form compounds, in practice one has to consider appropriate partial pressures and gas compositions ratios during deposition to ensure stoichiometric deposition from a metal target. This can be done by controlling the working pressure of the deposition chamber during sputtering and also by changing the flow rate of the gasses using a mass flow controller.

2.4 Determining film composition and density

Determining the composition dependent properties of the oxides hinges on accurate characterization of film composition. We relied on two methods: the first is to simply infer the composition profiles based on deposition rate calibrations, while the second is a direct measurement of the film composition using wave length dispersive spectroscopy (WDS).

Deposition rate calibrations were obtained by depositing elemental oxide films in the off-axis configuration on to a bare 3" Si wafer. Depositions were typically 20 minutes although in some cases, were extended to longer times if the deposition rate of the oxide was low. All depositions were eventually normalized to deposition rates (thickness/unit time) to account for the variability in deposition time. A bias of -75 V was applied to the substrate during calibration runs to emulate the deposition conditions of cosputtered

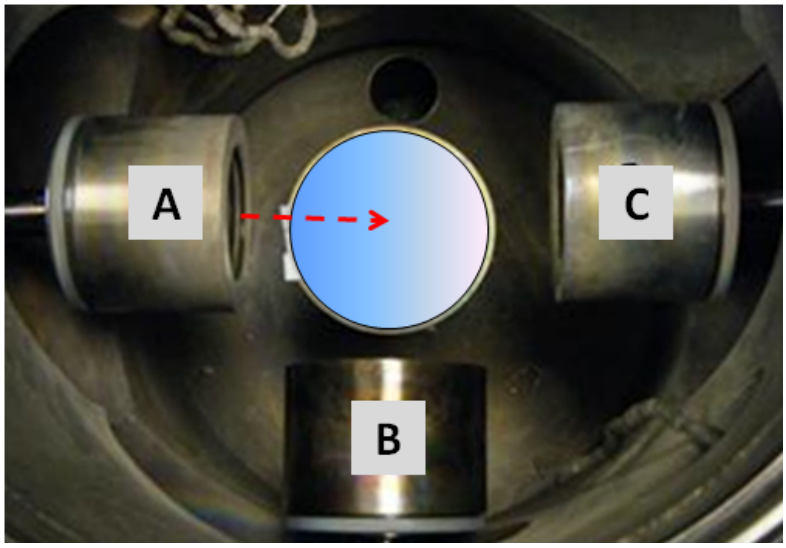


Figure 2.8: Guns in the off-axis configuration for deposition rate calibration. Figure depicts a deposition profile along Gun A.

films at room temperature.

The majority of the films studied were binary oxides thus requiring only linear thickness profiles across the diameter of the wafer (thickness measured using ellipsometry). This corresponds to a profile that starts from the center of the sputtering target and extends perpendicular to the target face (Figure 2.8). Three to four equally spaced points on the wafer that are collinear to this line were sufficient to generate an accurate linear thickness profile. An exponentially decaying function was fit to the measured thickness to obtain the decay length and the thickness at substrate edge. To determine the composition ratio of a binary oxide, we must first convert the thicknesses to molecules per unit area for each elemental oxide using the following equation,

$$[A_xO_y] = t(A_xO_y) * \text{A.M.}(A_xO_y) * \rho(A_xO_y) * N_a \tag{2.40}$$

For elemental oxide A_xO_y, $t(A_xO_y)$ denotes the thickness from the linear profile, A.M.(A_xO_y) is the atomic mass of the molecule, $\rho(A_xO_y)$ is the density of the molecule,

and N_a is Avogadro's number. Once the number of molecules per unit area was determined the composition ratio of molecule A_xO_y can be inferred by simply taking a ratio,

$$\frac{[A_xO_y]}{[B_pO_q] + [A_xO_y]} \tag{2.41}$$

in which the numerator represents the number of A_xO_y molecules per unit area, where as the denominator is the total number of molecules per unit area. The area term automatically cancels out since it is present in both the numerator and denominator. The ratio is only valid when the deposition times are consistent for each calibration.

Although composition profiles are typically highly reproducible (to ~1 at%), the conditions in the chamber may slightly change day to day affecting film properties between runs, so it is still instructive to perform a direct composition measurement. WDS is a sister technique to the EDS (energy-dispersive x-ray spectroscopy) and relies on the interaction of electrons with the material to stimulate emission of x-rays. The emitted x-ray energies are characteristic to the material and therefore in theory, the composition ratios can be discerned.

In comparison to EDS, WDS has a much higher spectral resolution; the resolution of EDS is typically on the order of ~ 100 eV, while WDS can resolve peaks down to several eV at full width half max (FWHM). [67] Improved resolution in WDS is achieved by preferential diffraction of characteristic x-rays through an analyzing crystal, in which the interaction of the x-rays with the crystal is dictated by Bragg's law, $n\lambda = 2d\sin(\theta)$. The detector, which collects the diffracted x-rays, and the crystal are simultaneously moved such that angle θ is varied by changing the position of the detector relative to the sample. Varying θ in turn selects which wavelength is diffracted according to Bragg's law.

WDS's accuracy is further improved by the use of standards with known composi-

tion. The task now is to take the measured intensity from the sample for element i, I_i, and compare it to the intensity measured from the standard $I_{(i)}$ to determine the composition in the sample. As a first approximation, one can determine the "k ratio,"

$$\frac{C_i}{C_{(i)}} = \frac{I_i}{I_{(i)}} = k_i \tag{2.42}$$

which assumes that a ratio between I_i and $I_{(i)}$ determines the mass composition ratio $C_i/C_{(i)}$, which are the composition fraction of the element of interest in the sample, and the composition fraction of the element of interest in the standard respectively. Complicating the analysis are "matrix effects," e.g. absorption and fluorescence, which have to be taken into account since they lead to errors in the k-ratio. Matrix effects are divided into three effects: atomic number (Z_i), x-ray absorption (A_i), and x-ray fluorescence (F_i). We can obtain the corrected k ratio by multiplying the correction factor for element i,

$$k(\text{corrected})_i = k_i[ZAF]_i \tag{2.43}$$

2.4.1 X-ray reflectivity

To calculate the polarizability of the material using the Clausius-Mossotti relation one needs to know the dielectric constant, ε_r, and density, ρ. While the dielectric constant is easily determined by C-D measurements, the density of an amorphous material is more difficult to ascertain, particularly since we require a systematic evaluation of ρ as a function of composition (position on the wafer). For our study, we rely on x-ray reflectivity (XRR) measurements to probe the density of the material at various points on the wafer.

X-ray reflectivity works by total external reflection, which occurs when x-rays are shined on a material with an angle of incident (θ) that is less than the critical angle θ_c

— where the critical angle is typically $\theta_c < 0.5$. Total external reflection only occurs because the refractive index, n, of a material is less than one at x-ray wavelengths. Because the condition for total external reflection depends on the electron density of the material, and therefore mass density, measuring the critical angle is an indirect measurement of the material density.

Reflectivity measurements were carried out using the Rigaku Smartlab diffraction system. Sample alignment is crucial to XRR measurements, hence we aligned the sample at each point on the wafer prior to each measurement. Standard 40 keV Cu K-α X-rays were conditioned through a slit assembly to minimize the beam footprint (footprint $\leq$ 10mm×3mm). Note, a large beam footprint is undesirable since we want to minimize the compositional averaging from the sample. In some cases a Ge (220)×2 monochrometer was utilized to further minimize the beam footprint. The reflected x-rays were further conditioned by the receiving slit assembly before being collected by the detector. To simplify the analysis, we took the critical angle to be the angle in which the intensity becomes half the total reflected intensity. [68] The critical angle is related to the mass density by the following formula

$$\rho = \frac{\theta_c^2 \pi A}{Z r_e \lambda^2 N_A} \tag{2.44}$$

where A is the atomic mass unit, r_e is the classical electron radius, λ is x-ray wavelength, N_A is Avogadro's number, and Z the atomic number. [69] Intrinsic to this equation is the assumption that $\theta_c = \sqrt{2\delta}$, which requires that the atomic form factor $f_1^0 \approx Z$ and that $f_2^0 << f_1^0$. These are fairly reasonable assumptions to make at these x-ray energies. [70]

2.5 The structure of non-crystalline materials

The Merriam-Webster Webster dictionary defines the word amorphous as "having no definite form," suggesting that amorphous materials lack any kind of ordering. However, this is far from the truth. In fact, amorphous materials consist of recurring structural elements that exhibit ordering in short range (typically several nearest neighbor distances). The complexity of the matter is evidenced for example by the on-going debate over the structural topology of amorphous Si. [71] The fact is, it merely appears that amorphous materials lack ordering because conventional methods that probe structure (e.g. x-ray diffraction using Cu-Kα radiation) are typically not able to uncover short or even medium range order. Therefore, researchers must rely on more specialized methods [72] to unravel the "hidden" structure of amorphous materials.

We rely heavily on several preexisting techniques that have experienced historic development, of which one is real space diffraction analysis. The goal of the technique is to determine a pair distribution function (PDF), $g(r)$, that represents the probability of finding an atom at distance r from an origin atom. Several references on this topic are available. [73, 74, 75, 76] To obtain $g(r)$ we must compute $\varphi(Q)$, the reduced intensity function from measured X-ray intensities, $I(Q)$, and then apply a sine Fourier transform. The canonical expression of the PDF is,

$$g(r) = \rho_a + \frac{1}{2\pi^2 r} \int_0^\infty \varphi(Q) \sin Qr \, dQ \tag{2.45}$$

Binary alloys require further mathematical treatment (to extract partial distribution functions for proper structural analysis), and ternary even more so. Recent improvements in computational power allow modeling of complex amorphous systems with great accuracy, [77, 45] and can be used in conjunction with experimental data to determine partial distribution functions. In the following section we derive the PDF for the simple case of a single element system, similar to the derivation found in [74].

In the kinematical limit the scattering intensity is written as the sum of amplitudes from different atoms multiplied by its complex conjugate. Defining a difference vector $\mathbf{r}_{ji} = \mathbf{r}_j - \mathbf{r}_i$:

$$I(Q) = \sum_j \sum_i f_j f_i e^{(2\pi i/\lambda)(\mathbf{s}-\mathbf{s}_0)\cdot\mathbf{r}_{ji}} \tag{2.46}$$

We can separate the sums for terms where $i = j$

$$I(Q) = \sum_j f^2 + \sum_j f^2 \sum_{i\neq j} e^{(2\pi i/\lambda)(\mathbf{s}-\mathbf{s}_0)\cdot\mathbf{r}_{ij}} \tag{2.47}$$

Introducing a density function $g_j(\mathbf{r}_{ij})$ we can write the the sum for the $i \neq j$ term as an integral over the volume element dV

$$I(Q) = \sum_j f^2 + \sum_j f^2 \int g_m(\mathbf{r}_{ij}) e^{(2\pi i/\lambda)(\mathbf{s}-\mathbf{s}_0)\cdot\mathbf{r}_{ij}} dV \tag{2.48}$$

Adding and subtracting the average atom density in the sample ρ_a, we obtain two integrals over the sample volume:

$$I(Q) = \sum_j f^2 + \sum_j f^2 \int [g_m(\mathbf{r}_{ij}) - \rho_a] e^{(2\pi i/\lambda)(\mathbf{s}-\mathbf{s}_0)\cdot\mathbf{r}_{ij}} dV + \sum_j f^2 \int \rho_a e^{(2\pi i/\lambda)(\mathbf{s}-\mathbf{s}_0)\cdot\mathbf{r}_{ij}} dV \tag{2.49}$$

To solve the integral we take an orientational average over the volume of the sample, assuming an isotropic distribution of atoms. The average density function is $g(\mathbf{r}) = \langle g_j(\mathbf{r}_{ij})\rangle$, in which $\mathbf{r} = \mathbf{r}_{ij}$ for fixed atom displacements. We consider amorphous materials of the type such that for large $\mathbf{r}$ values, $g(\mathbf{r}) - \rho_a$ goes to zero. We can thus simplify the sum of the self-scattering terms by replacing the sum with N. Also, the third term only contributes to I(Q) only at small scattering vectors, so it can be dropped from the equation if small angles are not included in the data. Substituting the scattering vector $Q = 4\pi \sin\theta/\lambda$ and from the fact that $(\mathbf{s} - \mathbf{s}_o)\cdot(\mathbf{r}_{nm}) = 2\sin\theta\, r\cos\phi$ we can solve the integral in equation 2.49

$$I(Q) = Nf^2 + Nf^2 \int_0^\infty \int_0^\pi [g(r) - \rho_a] e^{ikr\cos\phi} 2\pi r^2 \sin\phi \, d\phi \, dr \tag{2.50}$$

$$I(Q) = Nf^2 + Nf^2 \int_0^\infty 4\pi r^2 [g(r) - \rho_a] \frac{\sin kr}{kr}\, dr \tag{2.51}$$

To get the PDF we introduce the reduced intensity function $\varphi(Q)$

$$\varphi(Q) = \left[\frac{I(Q) - Nf^2(Q)}{Nf^2(Q)} \right] Q \tag{2.52}$$

which gives us,

$$\varphi(Q) = 4\pi \int_0^\infty r[g(r) - \rho_a] \sin kr dr \tag{2.53}$$

Currently phi is a function of Q-space. The PDF is a real space function and therefore, we need to apply a Fourier transform to $\varphi(Q)$. Defining $f(r) = r[g(r) - \rho_a]$, we obtain the following Fourier pair:

$$\begin{aligned} \varphi(Q) &= 4\pi \int_0^\infty f(r) \sin kr\, dr \\ f(Q) &= \frac{1}{2\pi^2} \int_0^\infty \varphi(Q) \sin kr\, dr \end{aligned} \tag{2.54}$$

From this we see that with a little rearranging, one can easily obtain the result presented in equation 2.45. Experimentally, the challenge lies in accurately measuring the scattered intensity $I(Q)$ and applying the proper correction algorithms such that an accurate PDF can be obtained. All diffraction spectra in this study were obtained at the Cornell High Energy Synchrotron Source (CHESS) A2 line at 40keV, corresponding to a x-ray wavelength of 0.030657 nm.

2.5.1 Silicon nitride membranes for x-ray diffraction

Silicon nitride membranes were designed and fabricated by Lauren Taylor

The signal from the silicon diffuse scattering adds an extra step in the data reduction process when characterizing thin films deposited on single crystal silicon in transmission. [78] For amorphous materials, this extra signal from silicon diffuse scattering is

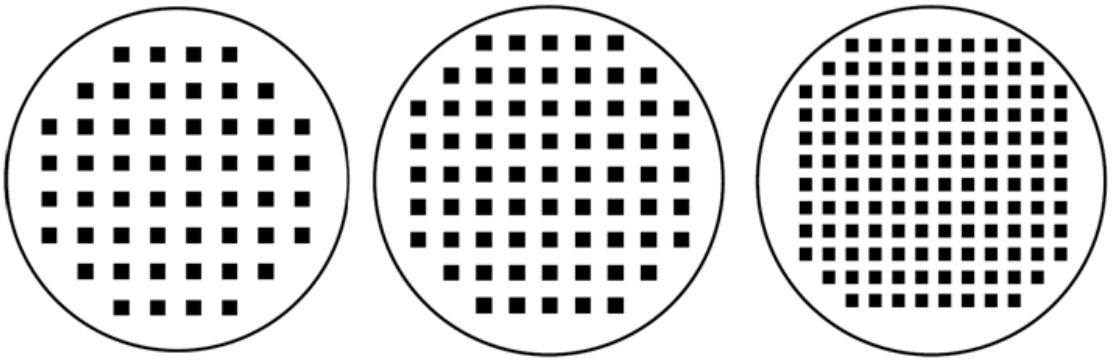


Figure 2.9: Cartoon showing the different silicon nitride membrane arrays used for the diffraction studies. From left to right: 8×8, 9×9, and 12×12 arrays.

especially problematic since the captured signal from the amorphous materials are not as intense in comparison to crystalline materials. To circumvent this problem altogether, we performed scattering experiments on films that were deposited on amorphous silicon nitride membranes. Because the nitride membranes are thin (a few microns thick) and amorphous, the subtraction process can be performed with greater ease.

Silicon nitride membrane arrays were fabricated using standard clean room processes in Cornell's Nanofabrication Facility (CNF). The membranes were arranged in a two-dimensional array as shown in Figure 2.9. Window size was modified depending on the number of membranes in the array. The windows for the 12×12 array were 2 mm on each side, whereas the windows for the 8×8 and 9×9 arrays were 3 mm.

Prior to silicon nitride deposition, 3 inch double-side polished wafers were cleaned in an acid bath followed by a base bath and rinsed in deionized water after each. The wafers were then transferred to a MRL Industries low pressure chemical vapor deposition (LPCVD) furnace where low stress silicon nitride was deposited. Nitride deposition rate was approximately 30 Å per minute determined from prior calibration runs. The substrates were kept in the furnace for 7 hours at 775 °C, which yielded nitride films that were approximately 1.5-1.6 microns thick determined by reflectometry (FilMetrics

Film Measurement System). Window array patterns were developed on the wafer using standard photolithographic methods. This involves transferring a photoresist pattern of the array on to the nitride coated substrate and etching away the nitride in the exposed windows. Finally, the wafer was back-etched using a potassium hydroxide solution forming the silicon nitride windows.

2.5.2 From measured intensity to PDF

Data acquisition is only half the battle, the other being data reduction and analysis. To refine the raw data into something we can analyze, we first normalized to the total x-ray flux (IC3 counts) and subtracted the silicon nitride background from the raw 2D spectra. The spectra was then and collapsed into a 1D diffraction pattern by integrating azimuthally in fit2D. [79]

At this point it is common to apply further corrections to the diffraction pattern. This includes polarization, absorption, multiple scattering effects etc. and have been widely implemented in several PDF analysis programs. [80] However, it is fairly time consuming to master these programs and require careful consideration of the various errors in diffraction analysis that call on a high level of expertise. In this study we make use of a less arduous approach by analyzing the data *ad hoc*.

The *ad hoc* approach is possible because of certain general properties of the structure function. Corrections to measured intensity, $I_m(Q)$, are expected to be either additive or multiplicative,

$$I_c(Q) = a(Q)I_m(Q) + b(Q) \tag{2.55}$$

where $a(Q)$ and $b(Q)$ are the Q dependent multiplicative and additive factors to the measured intensity, which if varied appropriately, gives us the correct form of the coherent

scattered intensity, $I_c(Q)$. The correction factors can also be related to total scattering structure function $F(Q)$ (as denoted by some researchers), essentially equivalent to $\varphi(Q)$ found in Equation 2.52

$$F(Q) = \alpha(Q)I_m(Q) + \beta(Q) \tag{2.56}$$

The critical insight is that although $\alpha(Q)$ and $\beta(Q)$ are not known *a priori*, the general form and behavior of the correction factors are known. Hence, the correct forms of $\alpha(Q)$ and $\beta(Q)$ can be determined variationally using PDFgetX3. [81]

We feed the background subtracted diffraction patterns into PDFgetX3 using the Python based command-line interpreter. [81] The program then variationally adjusts the correction factors and applies a sin Fourier transform to $F(Q)$ to solve for the PDF:

$$g(r) = \frac{2}{\pi} \int_{Q_{min}}^{Q_{max}} F(Q) \sin QrdQ \tag{2.57}$$

Note, that because we cannot collect data to infinite wavelengths, Q_{max} is always a finite value. This is equivalent to convolving a step function beyond Q_{max}, which becomes convolved with the data, resulting in "truncation" errors. In order to reduce this error, researchers typically collect data to sufficiently large Q values to incorporate large number of Fourier components to perform the transform. [82]

Direct information such as bond distance and coordination number are immediately available from the refined PDF. As mentioned earlier a more involved, but illuminating approach is simulating the amorphous structure. [83, 45] This enables us to not only confirm what we learn from the direct analysis, but also gives us additional information such as the partial distribution functions for multicomponent systems and electrical properties such as the dielectric constant.

2.5.3 X-ray absorption fine structure

The PDF method provides us with valuable local structural information. Nonetheless, the technique is not element specific. X-ray absorption fine structure (EXAFS, or XAFS) — a method which falls under X-ray absorption spectroscopy (XAS) — is an *element specific* local probe of the material, which can be used to extract information much like the PDF method and further, gather chemical information about the material such as the valence state of an element.

EXAFS capitalizes on the wave-particle behavior of electrons. Within the x-ray energies we are interested in ($1 \leq \mathrm{E} \leq 100$ keV) x-rays are absorbed by the photoelectric effect. Absorption of the x-ray excites an electron out of the atom leaving a core-hole. The element specificity is achieved by tuning the energy of the x-ray such that we only look at the absorption event for a specific element. Several events can also follow the absorption process such as the annihilation of the core-hole via fluorescence, or the emission of an Auger electron through an Auger process. The absorption of the x-ray results in the attenuation of the original x-ray intensity, I_o to I, which is measured by the detector. The relation between I_o and I can be expressed by the Beer-Lambert law

$$I = I_o e^{-\mu(E)t} \tag{2.58}$$

in which the absorption coefficient $\mu(E)$ is also an energy dependent quantity. The presence of neighboring atoms scatters the emitted photo-electron and in this process interferes with other scattered photo-electrons. This interference effect produces fringing in the absorption coefficient known as the "fine structure," which provides information about the local environment in the material. We can write the fine structure in terms of the absorption coefficients,

$$\chi(E) = \frac{\mu(E) - \mu_o(E)}{\Delta\mu_o(E_o)} \tag{2.59}$$

where $\mu_o(E)$ is the background absorption and $\Delta\mu_o(E_o)$ is the edge step which normalizes the equation to one absorption event. We can extract useful information from the data by modeling the fine structure. Taking into account inelastic and multiple scattering, and thermal vibrations the EXAFS equation is

$$\chi(k) = \sum_i \frac{N_i S_o^2 f_i e^{-2R_i/\lambda(k)} e^{-2k^2\sigma_i^2}}{kR_i^2} \sin 2kR_i + \delta_i(k) \tag{2.60}$$

Here, we have expressed the EXAFS as a function of the photo-electron wave number $k = \sqrt{\frac{2m(E-E_o)}{\hbar}}$. By fitting to the EXAFS equation, we can extract information such as bond distance and coordination number centered around a specific element.

All EXAFS experiments were carried out in CHESS's F3 line. We tuned the energy of the x-rays to the desired absorption edge (Ta L-3 absorption edge for tantalum based materials which sits at ~9883 eV). Ta metal foil was also used to calibrate the absorption edge. Data reduction and analysis was performed using the ATHENA and ARTEMIS software packages.[84]

CHAPTER 3

DIELECTRIC PROPERTIES OF AMORPHOUS ZR-AL-O AND ZR-SI-O THIN FILMS

This chapter is adapted from T. A. Naoi, Hanjong Paik, M. L. Green and R. B. van Dover: "Dielectric Properties of Amorphous Zr-Al-O and Zr-Si-O Thin Films" in *Journal of Advanced Dielectrics*.

3.1 Abstract

We have systematically studied the composition dependence of the dielectric properties of $Zr_{1-x}Al_xO_{2-x/2}$ and $Zr_{1-x}Si_xO_2$. An essentially linear variation of the static dielectric constant, ε_s, was observed as a function of composition, x, for compositions rich in the p-block element, i.e., $x > 0.4$, for both chemical systems. However an abrupt change in ε_s is found near $x \approx 0.35$, associated with the onset of crystallinity in as-deposited films. Breakdown fields do not show a comparable composition dependence. Measurements of the index of refraction at optical frequencies, combined with a simple Clausius-Mossotti interpretation, indicates that low-frequency (ionic) contributions to the polarizability exhibit systematic deviation with respect to values linearly interpolated from the endmembers. These trends are not consistently affected by the presence of crystalline order, but are related to changes associated with heterogeneous local oxygen coordination and bonding.

3.2 Introduction

Amorphous dielectrics comprising mixed transition-metal/p-block (e.g., Hf, Zr or Ta combined with Al, Si or Ge) oxides are technologically important and are generally viewed as following simple trends as a function of composition. Some examples (e.g., Zr-Si-O) have been found to exhibit unexpected behavior, such as an enhanced dielectric constant. [25, 3, 37, 47] Amorphous HfO_2-based dielectrics, modified with the addition of SiO_2 and/or Al_2O_3, are now widely used in commercial integrated circuits as alternatives to the native SiO_2 dielectric that controls charge transport in Si-channel field-effect transistors, because HfO_2, like other oxides of the Group 3, 4, and 5 elements, has a significantly higher dielectric constant (static value of the relative dielectric constant, $\varepsilon_s = \varepsilon(\omega \rightarrow 0)/\varepsilon_0 \approx 23$) than SiO_2 for which $\varepsilon_s \approx 3.9$. Compared to SiO_2 they can therefore achieve a comparable charge density at a specified gate voltage using a layer thick enough to avoid excessive leakage by direct tunneling. [20] Ab-initio studies of amorphous $Zr_{1-x}Si_xO_2$ and $Hf_{1-x}Si_xO_2$ and related materials have highlighted the importance of local structure on dielectric properties, and the consequent impact on the composition dependence. [28, 85, 27, 86]

In order to understand the behavior of this interesting class of materials, we have focused on the Zr-Al-O and Zr-Si-O systems. Films deposited at room temperature exhibit a clear transition from nanocrystalline (Zr-rich compositions) to amorphous (Si- or Al-rich compositions). Polycrystalline and amorphous thin films of these materials exhibit comparable values for ε_s; in both case the value represents an approximate directional average of the anisotropic value associated with single crystals. The question of crystallinity can be important; for example, as gate dielectrics, amorphous oxides are advantageous over polycrystalline materials since they avoid inhomogeneity in local charge density due to anisotropy that would degrade mobility in the conducting chan-

nel of an FET. Polycrystalline dielectrics are expected to exhibit a lower breakdown field than that of amorphous dielectrics due to the presence of grain boundaries. [87] We are using these two systems to determine the effect of the presence or absence of crystallinity on the dielectric constant and the breakdown field, and to determine the systematic effect of the Zr/Si or Zr/Al ratio on these properties. This information provides insight regarding the mechanism for the observed enhancement of the dielectric response.

We used a composition-spread technique for thin film growth to enable composition-dependent trends to be identified with low composition-to-composition process variation and very fine composition resolution. The dielectric constant of amorphous films was found to vary roughly linearly with composition, while the static dielectric constant of crystalline (transition-metal-rich) films was found to be significantly higher. Leakage currents and breakdown fields were less strongly composition dependent and suggest that grain boundaries in nanocrystalline samples do not serve as weak links.

3.3 Experimental Methods

Compositionally-graded films were prepared using 90° off-axis reactive cosputtering using two 2-inch planar magnetron sputter guns arranged in a facing-targets configuration, as described previously. [88] The 75 mm diameter Si substrates were positioned between the two guns so that species sputtered from each target condense on the substrate simultaneously. The deposition rate decreases rapidly as a function of distance from the target, so a natural composition gradient is obtained in this geometry. [88] The sputter guns were operated at rf power levels chosen to achieve approximately 1:1 cation ratio at the center of the substrate. The films were about 100 nm thick with cations randomly

mixed on an atomic scale. Films used for evaluation of crystallinity were deposited on bare Si wafers while those used for evaluation of electrical and optical properties were deposited on wafers pre-coated with sputtered TiN). Prior to film deposition the vacuum chamber was evacuated to a pressure of 10^{-6} Torr or lower. Gas flow rates during deposition were 15 std. cm^3/minute (sccm) Ar and 10 sccm O_2, with the turbopump throttled to maintain a pressure of 30 mTorr in the chamber. The substrate was also intentionally biased with 10 W rf power, yielding a dc potential of -75 V that ensures formation of a dense film (avoiding formation of columnar voids during deposition. [61]) The substrate temperature during deposition was maintained below 60 °C by attaching the substrate to a heatsink. Under these conditions the deposition rate was roughly 0.1 nm/s,though the rate varies somewhat with position on the sample. The final thickness of the films was on the order of 100 nm.

The composition of the films intentionally varies along the length of the sample. Quantitative composition variation was determined using calibration samples on which a single element was reactively sputtered under the same conditions used for the cosputtered spreads. The thickness of the deposited film, measured as a function of position using a profilometer, was used to calculate the deposition rate. The cation composition of the cosputtered sample is then inferred simply by calculating the ratio of arrival rates of the two cation species. The oxygen stoichiometry in these systems was calculated using charge balance assuming cations are in their highest oxidation state, an assumption that is validated by the observation of low leakage currents in the dielectric. An example of the position dependence of composition is shown in Fig. 3.1. At selected positions the codeposited samples were examined using Rutherford Backscattering Spectroscopy (RBS). The RBS analysis, which has an absolute accuracy estimated to be better than 5%, yielded compositions that were consistent with the results of the rate calibrations, confirming the validity of all of the assumptions inherent in the latter technique. A

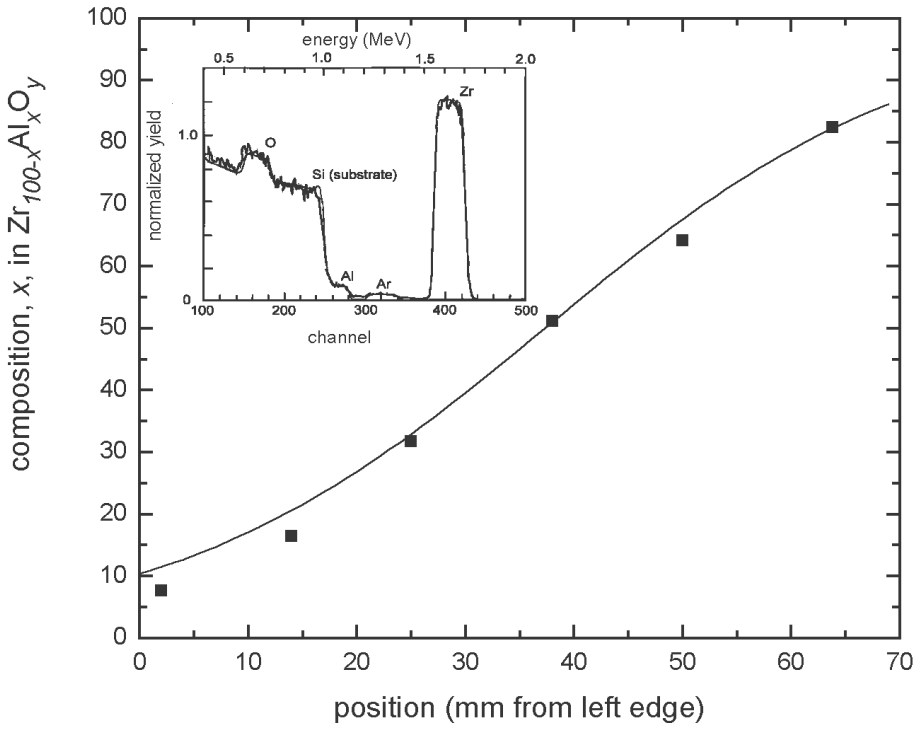


Figure 3.1: Composition as a function of position along the 7-cm sample axis. Solid line: compositions calculated using rate calibrations for each species (ZrO_2, Al_2O_3). Squares: Compositions inferred using Rutherford Backscattering Spectroscopy. Inset: a typical RBS spectrum used to determine the composition. A small level of Ar incorporation is typical in sputtered films, and can be seen in the RBS spectrum, though it has a negligible effect on electrical or structural properties.

typical RBS spectrum is shown in the inset to Fig. 3.1.

The thickness of the films is not strictly constant across the length of the sample, since the sum of the deposition rates from the two targets is not constant. The thickness was therefore determined as a function of position by preparing a codeposited sample with a liftoff mask. The mask was then removed and the thickness of the deposited film was determined at fifteen positions using a profilometer. These measurements of the dependence of thickness with position were interpolated using a quadratic fit.

X-ray diffraction patterns were taken with a Brucker-AXS system equipped with a Cu K_α generator and crossed-wire area (two-dimensional) detector. The resulting data were integrated over the range -47° < χ < -133° to yield a single graph of integrated intensity versus 2θ. The crystallites exhibited no preferred orientation. Fig. 3.2 illustrates the reflections associated with monoclinic ZrO_2, observed for Zr-rich compositions that exhibit nanoscale crystalline order. The width of the peaks, corrected for instrument resolution, is a measure of the crystallite size (structural coherence length, L) as inferred using the Debye-Scherrer formula, $L = 0.6\lambda/Bcos\theta$, where B is the full width at half maximum. The pattern corresponding to $Zr_{0.75}Si_{0.25}O_2$ is marked with the (hkl) Miller indices of the monoclinic phase; the distinct peak expected for cubic or tetragonal ZrO_2 at approximately 30° is clearly not present in the pattern. The pattern corresponding to $Zr_{0.89}Al_{0.11}O_{1.94}$ is less well defined, but is consistent with monoclinic ZrO_2. It is surprising that the monoclinic phase should be present in nanocrystalline $Zr_{0.89}Al_{0.11}O_{1.94}$, both because incorporation of a trivalent ion in ZrO_2 introduces O vacancies that typically stabilize the cubic fluorite phase, as in yttrium-stabilized zirconia, and because grain refinement in unsubstituted ZrO_2 also stabilizes the cubic phase. [89] When an identical $Zr_{1-x}Al_xO_{2-x/2}$ composition spread was annealed at 1000 °C for 20 seconds in N_2, Zr-rich compositions ($x < 0.18$) exhibited a mixture of monoclinic and tetragonal phases while for $0.18 < x < 0.40$ the structure was entirely cubic, a result that is not inconsistent with expected equilibrium behavior. For $x > 0.4$ a mixture of cubic and amorphous material was present. Annealing $Zr_{1-x}Si_xO_2$ at 1000°C yielded surprisingly similar results, considering that tetravalent Si will not introduce O vacancies to maintain charge balance. Zr-rich compositions ($x < 0.35$) exhibited a mixture of monoclinic and tetragonal phases while for $0.35 < x < 0.80$ the structure is entirely cubic. Apparently introduction of Si, which favors tetrahedral coordination with O (rather than the octahedral coordination of ZrO_2), disrupts the interactions that stabilize the monoclinic

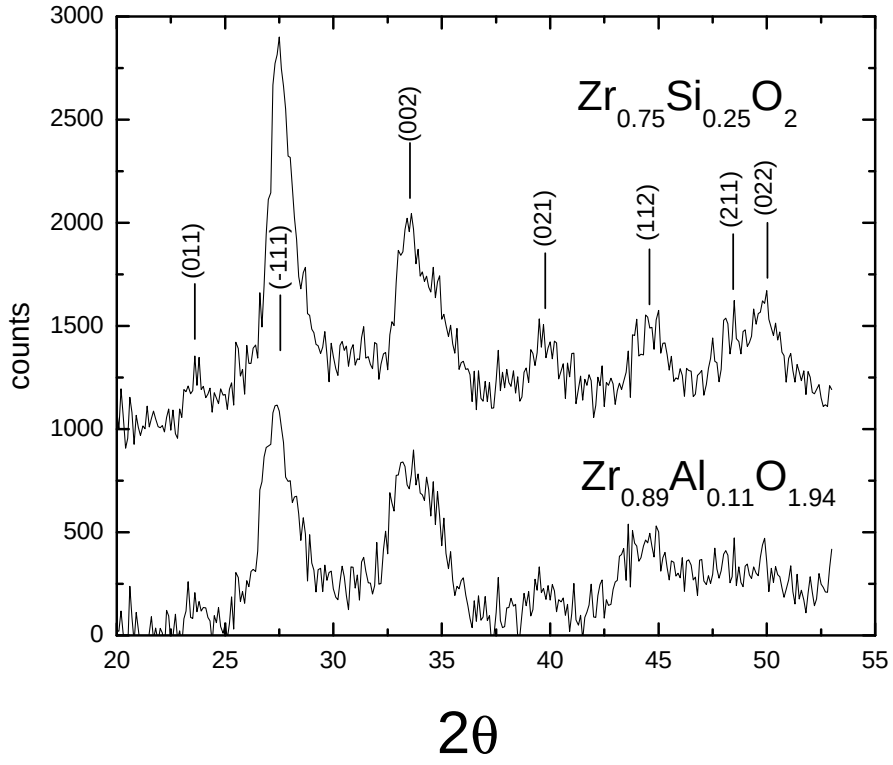


Figure 3.2: X-ray diffraction patterns for typical nanocrystalline Zr-rich samples. Top curve: $Zr_{0.75}Si_{0.25}O_2$ film as deposited, offset vertically for clarity, with positions for the strongest monoclinic reflections indicated. Bottom curve: $Zr_{0.89}Al_{0.11}O_{1.94}$, illustrating x-ray peaks that are broader due to the lower degree of crystallinity compared to the Zr-Si-O sample.

structure. Interestingly, no evidence for formation of zircon, $ZrSiO_4$, is observed in the annealed films. [24] The index of refraction and thickness of the films were determined using a Sentech SE801 spectroscopic ellipsometer over a wavelength range of 450-800 nm. The optical constants of the substrate layer were determined prior to film deposition and were fixed during the fitting process. A film/substrate bilayer model was used to fit ψ and δ, and the optical constants of the film were modeled using the Cauchy relation $n(\lambda) = n_0 + n_1/\lambda^2 + n_2/\lambda^4$ using a least-squares routine. The reported refractive index n was taken to be n_0, i.e., the infinite wavelength limit far from absorption edge.

The static dielectric constant, ε_s was evaluated using parallel-plate capacitors fabricated on oxide films grown on a non-reactive TiN base electrode. Pt counterelectrodes 20 nm thick and 200 μm in diameter were deposited on the films using electron-beam evaporation through a shadow mask. The capacitance of these structures was measured at 10 kHz using an automated probing station and an HP4192A capacitance meter. Some capacitors were measured from 100 Hz to 1 MHz, and were found to exhibit little dispersion over this range. The dissipation factor for all compositions was found to be low ($\tan\delta < 0.05$), indicating high quality films with no anomalous polarization modes. [61] Typically 10 to 40 capacitors were measured at each composition in the linear composition spread; the median value was chosen as representative. Current-voltage characteristics were also measured on the Pt/dielectric/TiN capacitor structures. A Keithley 617 source/electrometer was used to apply a stepped voltage across the capacitor and to measure the resulting leakage current. Fields from 0 to 10 MV/cm were applied (Pt counterelectrode biased positively), and current densities in the range $10^{-7} - 10^{-3} A/cm^2$ were measured. The breakdown field, E_{br}, was defined as the field at which the current density increased by a factor of 100 or more in a single 0.1 MV/cm step. In cases where the I-V curve was continuous even up to $10^{-3} A/cm^2$ (soft breakdown), the breakdown criterion was chosen to simply be the field at which the current density passed $10^{-4} A/cm^2$. Again, 10 to 40 capacitors were measured at each composition; because random defects are unavoidable in films fabricated and processed in a non-cleanroom facility we chose the maximum measured E_{br} as the best representative of the intrinsic behavior of the dielectric.

3.4 RESULTS AND DISCUSSION

3.4.1 Dielectric constant and refractive index

The composition dependence of the dielectric constant of $Zr_xAl_{1-x}O_{1.5+x/2}$ is presented in Fig. 3.3. Marks at $x = 0$ and $x = 1.0$ represent expected values for the endmembers. The dielectric constant of Al_2O_3 ($\varepsilon_s \approx 9$) is not controversial. [90] The dielectric constant of monoclinic ZrO_2 appears to be $\varepsilon_s \approx 32$ based on extrapolation of data measured on polycrystalline ceramic $(Zr,Y)O_x$ as reported by Henn, et al. [91]. We note that another reference suggests $\varepsilon_s \approx 17$ based on interpretation of infrared spectroscopy [92] and supported by first-principles calculations [93]. We note also that these calculations predict that metastable phases of ZrO_2 would have distinctly different values for ε_s, namely $\varepsilon_s \approx 37$ and $\varepsilon_s \approx 47$ for the cubic and monoclinic phases [93]. Other *ab initio* calculations [94] predict a value $\varepsilon \approx 21$ for amorphous ZrO_2 which we have marked on the left-hand axis.

The dielectric constant for amorphous (Al-rich) compositions follows a nearly linear interpolation between the values for the amorphous endmembers. A clear jump in ε_s is seen at approximately $x = 0.35$. The transition is associated with the change in crystallinity, as suggested by the top panel of Fig. 3.3, which reports the XRD coherence length (grain size) as a function of composition: the as-deposited films are completely amorphous for $x \gtrsim 0.35$. Extrapolating data for $x < 0.30$ to the left-hand axis yields a value in good agreement with that inferred for monoclinic ZrO_2.

The $Zr_{1-x}Si_xO_2$ system exhibits a similar behavior as illustrated in Fig. 3.4. Although the dielectric constant exhibits a nearly linear trend with composition for amorphous compositions ($0.35 < x < 0.90$) and the trend extrapolates to the measured value

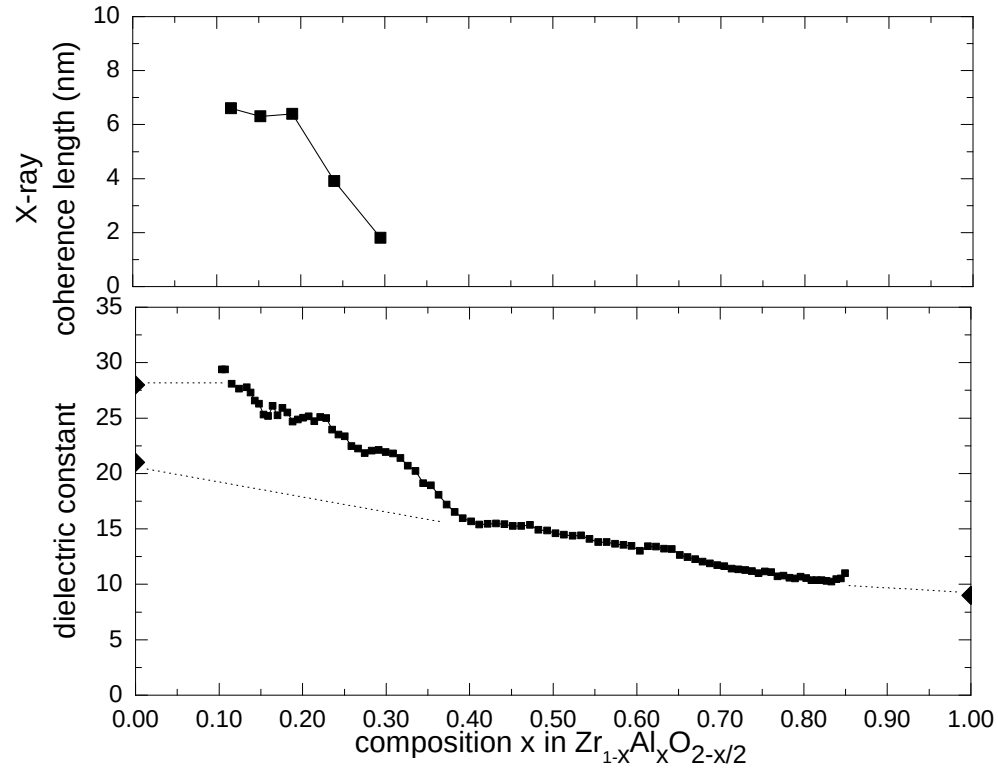


Figure 3.3: Bottom: Composition dependence of the dielectric constant of thin film $Zr_{1-x}Al_xO_{1.5+x/2}$ deposited at room temperature. Dotted lines represent extrapolations to values for the endmembers (crystalline and amorphous ZrO_2 and amorphous Al_2O_3) indicated with triangles.Top: composition dependence of the structural coherence length (i.e., essentially grain size) as inferred from X-ray diffraction data using the Debye-Scherrer formula.

for SiO_2 ($\varepsilon_s = 3.9$), for $x = 0$ it extrapolates to $\varepsilon_s = 24$ rather than the measured value for amorphous ZrO_2. As in the Zr-Al-O system the abrupt transition in the trend of ε_s with composition at $x \approx 0.35$ is associated with the change from crystalline to amorphous structure (Fig. 3.4, top panel). At $x = 0$ the extrapolation of the trend exhibited by crystalline compositions of both Zr-Si-O and Zr-Al-O yields the same value, $\varepsilon_s = 30$, for both, suggesting a common origin. However, we note that the crystal structure observed in these films is not the cubic structure bulk ZrO_2 described by Scott, which also extrapolates to $\varepsilon_s = 30$ for undoped ZrO_2, perhaps coincidentally.

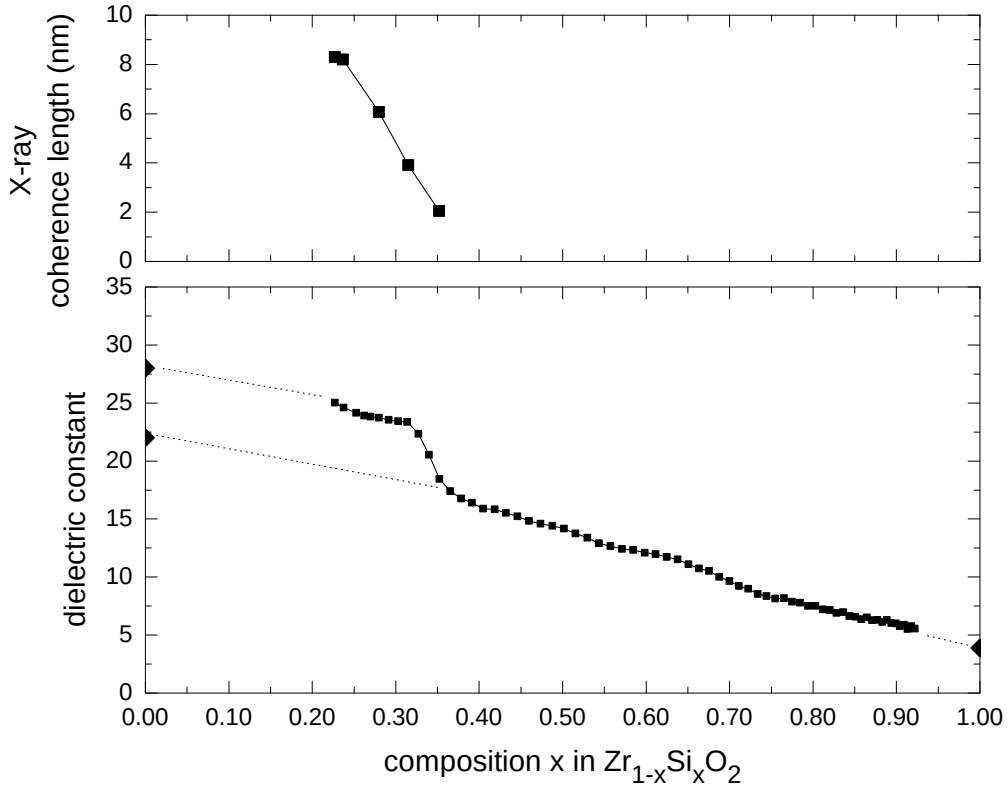


Figure 3.4: Bottom: Composition dependence of the dielectric constant of thin film $Zr_{1-x}Si_xO_2$ deposited at room temperature. Dotted lines represent extrapolations to values for the endmembers (crystalline and amorphous ZrO_2 and amorphous SiO_2) indicated with triangles.Top: composition dependence of the structural coherence length (i.e., essentially grain size) as inferred from X-ray diffraction data using the Debye-Scherrer formula.

The major trend in dielectric constant as a function of composition in these systems is undoubtedly due to a change in the polarizability from the small values associated with SiO_2 or Al_2O_3 to the large value associated with ZrO_2. But it is not immediately evident what fraction of the low-frequency polarizability can be attributed to ionic motion as opposed to polarization of the electrons with respect to atom cores, nor to what degree the density of the film plays a role. Fig. 3.5 presents the composition dependence of the index of refraction in these systems. The data for $Zr_{1-x}Al_xO_{2-x/2}$ follow a smooth trend that extrapolates almost linearly to accepted values for the endmembers. [93, 95]

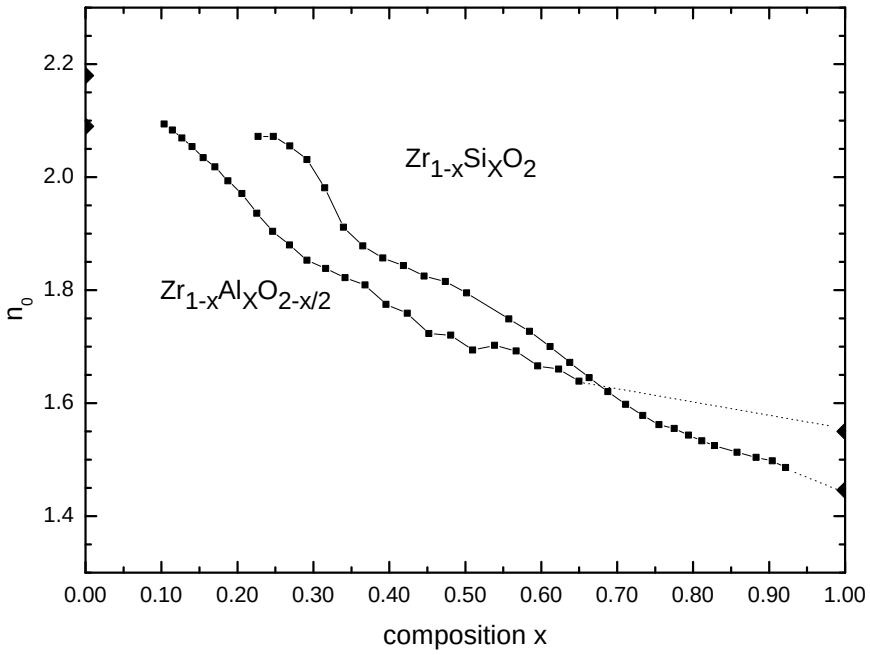


Figure 3.5: Bottom: Composition dependence of the refractive index n_0 of thin film $Zr_{1-x}Al_xO_{2-x/2}$ and $Zr_xSi_{1-x}O_2$ deposited at room temperature. Dotted lines represent extrapolations to values for the endmembers (crystalline and amorphous ZrO_2 and amorphous SiO_2 and Al_2O_3) indicated with triangles.

The data for $Zr_{1-x}Si_xO_2$ exhibits a more dramatic variation, extrapolating linearly to the endmember value for amorphous SiO_2. The data are insufficient to allow unambiguous extrapolation to ZrO_2, though they are consistent with the same value as inferred for $Zr_{1-x}Al_xO_{2-x/2}$. Where the data for Zr-Al-O show only a break in slope associated with the crystalline-amorphous transition, the data for Zr-Si-O exhibit a jump similar to that observed in the static dielectric constant.

3.4.2 Microscopic picture: static and electronic polarizability

The high frequency dielectric constant is simply $\varepsilon_\infty = n^2$. See supplementary material at [URL will be inserted by AIP] for graphs comparing composition dependence of the static dielectric constant with the ε_∞. But comparing ε_s with ε_∞ does not offer insight into the behavior at the local level because it does not account for depolarization effects and therefore does not account for density variations correctly. Interpreting local polarization behavior in this context does not mean associating a fixed polarizability with specific ions, but is simply a way to compare trends. A baseline expectation might be that the composition dependence of the polarizability interpolates linearly between the values for the endmembers. Deviations from that trend would suggest unexpected interactions or structural anomalies.

The Clausius-Mossotti equation relates the local polarizability to the externally observed dielectric constant:

$$\frac{\varepsilon_s - 1}{\varepsilon_s + 2} = \frac{4\pi}{3} \sum_m \frac{\alpha_j}{V_m}$$

where α_j is the polarizability due to mechanism $j \in \{electronic, ionic, dipolar, \ldots\}$, and V_m is the molar volume. For the materials under consideration only electronic (α_e) and ionic (α_{ion}) mechanisms are present. At optical frequencies, well above any phonon modes, only the electronic mechanism is active, so we can infer α_e/V directly from the refractive index. These values are plotted in Figs. 3.6 and 3.7, for Zr-Si-O and Zr-Al-O, respectively.

The data for SiO_2-rich Zr-Si-O closely follow a simple linear interpolation between the values for the endmembers; the slight deviations are not thought to be meaningful. For $x < 0.35$ the deviation is significant and is associated with the presence of nanocrystallity. The data for $0.20 < x < 0.30$ are nearly constant, and match the value for unsubstituted ZrO_2. We speculate that introduction of SiO_2 decreases the polariz-

ability and the molar volume at compensating rates, leaving α_e/V constant, and that at $x \approx 0.35$ the transition to a disordered structure leads to a fairly abrupt decrease in the molar volume.

The optical polarizability in the Zr-Al-O system follows a significantly different pattern. Here the trend deviates *below* the linear interpolation of endmember values, suggesting either an increased molar volume or an electronic interaction that lowers the polarizability of the system. The data follow a relative smooth curve and extrapolate accurately to the values for the endmembers. In the Zr-Al-O system there is no abrupt transition in α_e associated with the crystalline-amorphous structural transition.

Figures 3.6 and 3.7 also show the composition dependence of the static polarizability, α_s/V, to which both electronic and ionic mechanisms contribute. In the $Zr_{1-x}Si_xO_2$ system, α_s/V is nearly constant with increasing SiO_2 fraction until at the crystalline/amorphous there is an abrupt decrease, presumably associated with an increase in molar volume. For $x > 0.35$ the trend of $\alpha_s(x)/V$ shows clear negative curvature, leading to a rapid decrease for the most SiO_2-rich compositions. In the $Zr_{1-x}Al_xO_{2-x/2}$ system a less pronounced decrease is associated with the crystalline/amorphous transition, and there is almost no curvature in $\alpha_s(x)/V$ for $x > 0.35$. Fairly abrupt transitions in both the optical and static polarizability are seen around $x = 0.3$ for both chemical systems, clearly associated with the transitions in the dielectric constant. These might be related to changes in molecular density, or could indicate a change in one of the mechanisms for the polarizability.

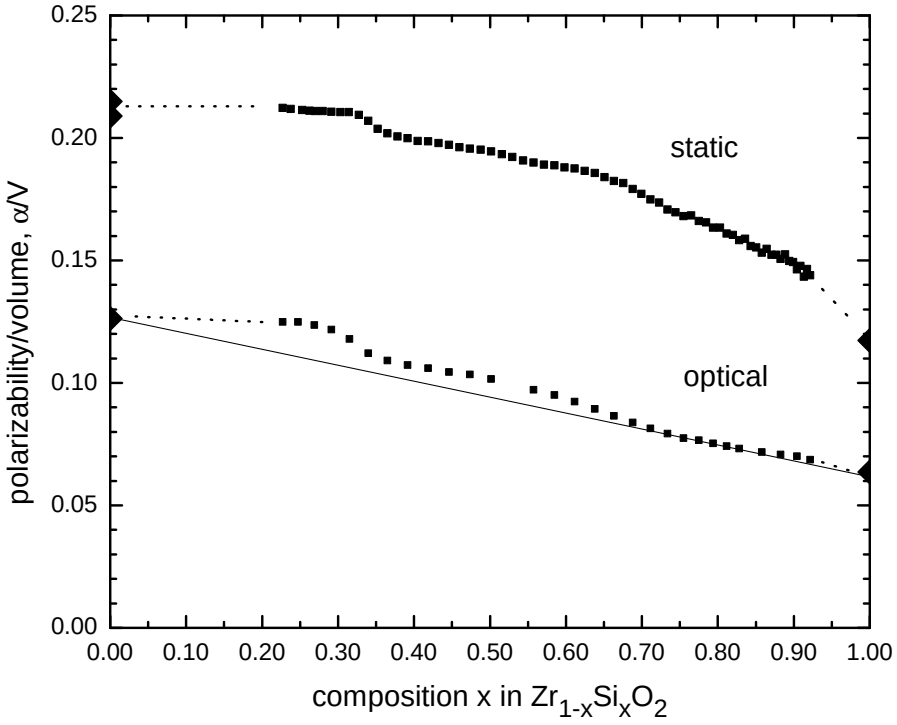


Figure 3.6: Composition dependence of the polarizability per unit volume, α/V. Dotted lines are extrapolations to values for the endmembers (crystalline and amorphous ZrO_2 and amorphous SiO_2) indicated with triangles. Top curve: polarizability inferred from the static (10 kHz) dielectric constant. Bottom curve: polarizability inferred from the optical index of refraction. The solid line is a simple interpolation between the values for the endmembers.

3.4.3 Microscopic picture; ionic polarizability

The difference between the static and optical dielectric responses is due to the displacement of ions, a mechanism that is active only at phonon frequencies and below. Thus the ionic contribution can be inferred using $\alpha_{ionic}/V = \alpha_s/V - \alpha_e/V$. In Fig. 3.8 we calculate this difference using the data of Figs. 3.6 and 3.7, in which the optical polarizability was interpolated using a cubic spline before subtracting from the static polarizability. Although the data exhibit small wiggles that are probably not meaningful, the two sys-

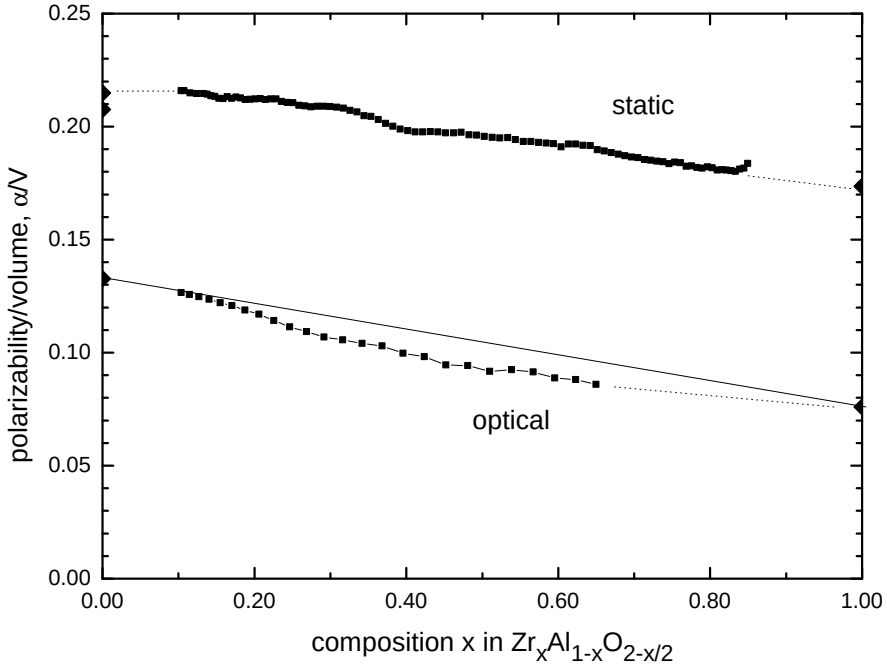


Figure 3.7: Composition dependence of the polarizability per unit volume, α/V. Dotted lines are extrapolations to values for the endmembers (crystalline and amorphous ZrO_2 and amorphous Al_2O_3) indicated with triangles. Top curve: polarizability inferred from the static (10 kHz) dielectric constant. Bottom curve: polarizability inferred from the optical index of refraction. The solid line is a simple interpolation between the values for the endmembers.

tems demonstrate clearly different trends. The data for both compositions spreads can be seen to extrapolate plausibly to standard values inferred for the endmembers. The Zr-Al-O system exhibits a distinct increase in $\alpha_{ionic}/V(x)$ for increasing x up to $x \approx 0.3$, after which α_{ionic}/V is nearly independent of x. The Zr-Si-O system shows almost the opposite trend: a roughly constant value for $\alpha_{ionic}(x)/V$ for $x < 0.7$ followed by a rapid decrease as the SiO_2 is increased further.

At $x = 0.3$, the data $\alpha_{ionic}(x)/V$ in Zr-Al-O do not exhibit a transition comparable to that seen in $\alpha_e(x)/V$; only a modest change of overall slope is observed. If the transition in $\alpha_e(x)V$ were due to a change in molecular volume V, a comparable transition should

be observed $\alpha_{ionic}(x)/V$. The lack of such a feature implies that the change is in the polarizability $\alpha_e(x)$ at the crystalline/amorphous transition is due to a reorganization of the electron density in the system. By comparison, the data for Zr-Si-O are consistent with a model in which the transition is due to a change in molecular volume V rather than α_{ionic} or α_e.

The composition dependence of the dielectric properties of Zr-Al-O and Zr-Si-O follow trends that are subtly but profoundly different, though this is not readily apparent in observations of only the static dielectric constant. The main differences are ascribed to substantial changes in bonding and ion dynamics associated with incorporation of ZrO_2 in amorphous SiO_2-rich Zr-Si-O and incorporation of Al_2O_3 in ZrO_2-rich Zr-Al-O (but not with incorporation of SiO_2 in ZrO_2-rich Zr-Si-O or incorporation of ZrO_2 in amorphous Al_2O_3-rich Zr-Al-O). We suggest that this reflects, in the first case, the difference in oxygen coordination for Si (typically tetrahedral) compared to that for Zr (typically octahedral). In the second case, it may also reflect the different valence of Al (typically 3^+) compared to that of Zr (typically 4^+), which would introduce oxygen vacancies to maintain charge neutrality. The role played by changes in molar volume are not determined in this study, but we speculate that the overall smooth trend in enhancement of the ionic polarizability in Zr-Al-O may be attributed to modest changes in molar volume. The abrupt change in the polarizability in Zr-Si-O at ≈35% Si can plausibly be attributed to a change in molar volume associated with the crystalline/amorphous transition. A more conclusive interpretation of the bonding changes would require data from local probes such as EXAFS, or Raman or infrared spectroscopy, coupled with *ab initio* calculations of average structure in the amorphous materials.

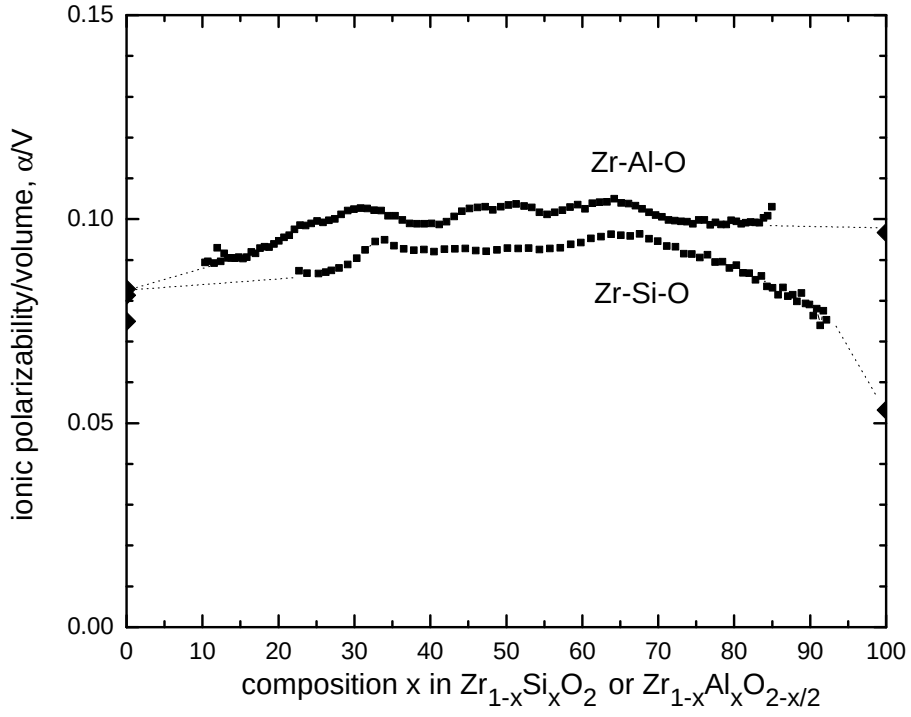


Figure 3.8: Composition dependence of the ionic polarizability,, as inferred from the text. Dotted lines represent extrapolations to values for the endmembers (crystalline and amorphous ZrO_2 and amorphous SiO_2 and Al_2O_3) indicated with triangles.

3.4.4 Breakdown field

The leakage current and breakdown field of these materials is of significant technological interest, as they affect their suitability for any application in which charge or energy storage is important. Both properties depend on extrinsic mechanism such as details of impurity concentration, local inhomogeneity, film smoothness, electrode composition, etc., so that few general conclusions can be drawn. Fig. 3.9 presents composition dependence of the breakdown field for $Zr_{1-x}Al_xO_{2-x/2}$ and $Zr_{1-x}Si_xO_2$. For both materials, the breakdown fields range from $\approx$ 4 MV/cm for the Zr-rich compositions to $\approx$ 6 MV/cm for the Al-rich and Si-rich compositions. The significant feature in this

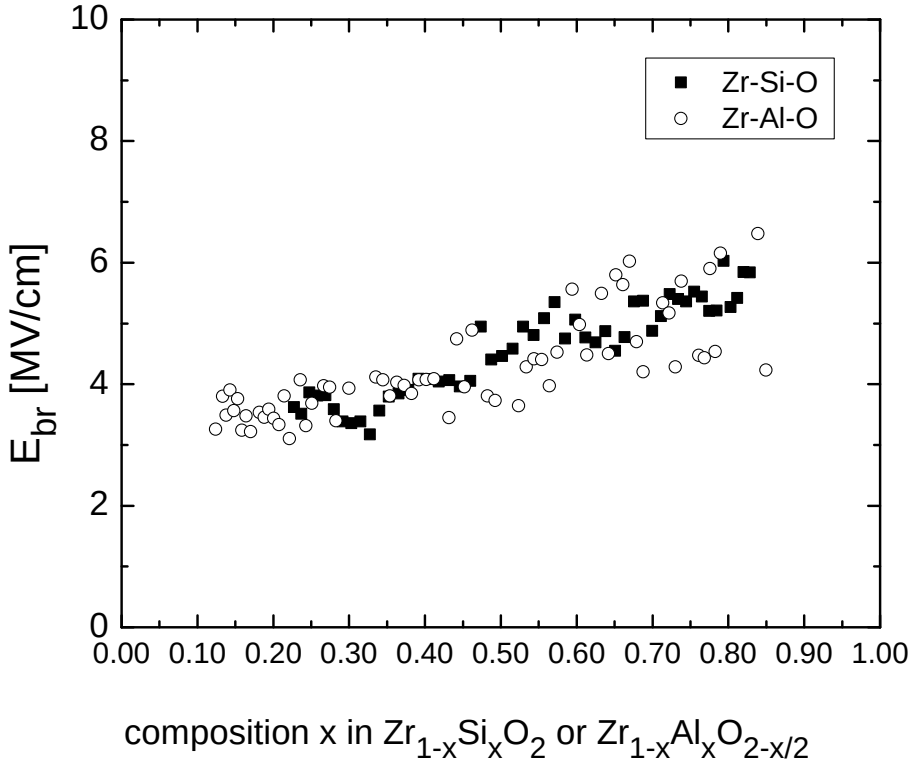


Figure 3.9: Composition dependence of the breakdown field in Zr-Si-O and Zr-Al-O. Solid symbols represent Zr-Al-O capacitors and open circles represent Zr-Si-O capacitors. Notice that no change in the trend is associated with the crystalline/amorphous transition at $x \approx 0.35$.

data is that in both systems there is no change in breakdown field associated with the crystalline/amorphous transition at $x \approx 0.35$. This result clearly repudiates the conventional wisdom that the breakdown field of thin film polycrystalline materials must be compromised due to "weak" grain boundaries, as is observed in bulk ceramics. As indicated by the crystallite size plotted in Figs. 3.3 and 3.4, the films have a high density of grain boundaries. This behavior suggests that grain boundaries in the thin films do not act as sinks for impurities and defects during growth, unlike those in high-temperature processed bulk ceramics.

3.5 Conclusion

This work shows that thin films of mixed transition-metal/p-block oxides, such as Zr-Si-O, are not as simple as they appear at first glance. Enhancement of the composition-dependent polarizability in amorphous SiO_2-rich Zr-Si-O is definitively confirmed in the data, while a comparable enhancement in Al_2O_3-rich Zr-Al-O is ruled out though an enhanced polarizability in ZrO_2-rich Zr-Al-O is observed. In Zr-Si-O, but not Zr-Al-O, a change in molecular volume appears to be associated with the crystalline/amorphous transition. The changes in low-frequency polarizability are most plausibly attributed to changes in ion dynamics (phonon spectrum), suggesting further experiments directly probing the dynamics by IR spectroscopy. No significant change in the breakdown field associated with the crystalline vs. amorphous condition.

3.6 Acknowledgments

This work was supported by the U.S. Department of Energy under Grant No. DE-FG02-06ER06-15. This work was performed in part at the Cornell NanoScale Facility, a member of the National Nanotechnology Infrastructure Network, which is supported by the National Science Foundation under Grant No. ECS-0335765. Additionally, this work made use of the Cornell Center for Materials Research facilities with support from the National Science Foundation Materials Research Science and Engineering Center program under Grant No. DMR-0520404. Preliminary data was obtained at Lucent Technologies Bell Labs. The authors thank Dr. R. M. Fleming for collecting and interpreting the RBS data.

CHAPTER 4

RELATION BETWEEN DIELECTRIC PROPERTIES AND STRUCTURE IN COMPLEX AMORPHOUS AND CRYSTALLINE THIN FILM OXIDES

This chapter is adapted from T. A. Naoi and R. B. van Dover: "Relation Between Dielectric Properties and Structure in Complex Amorphous and Crystalline Thin Film Oxides" in preparation.

4.1 Abstract

We have used off-axis magnetron sputtering to explore changes in the structural and dielectric properties of pseudo-ternary thin film dielectrics as a function of composition. We investigated four different oxide systems: $(HfO_2)_{1-x}(SiO_2)_x$, $(HfO_2)_{1-x}(GeO_2)_x$, $(Nb_2O_5)_{1-x}(GeO_2)_x$, and $(Ta_2O_5)_{1-x}(SnO_2)_x$. The diffraction patterns of $(HfO_2)_{1-x}(SiO_2)_x$ and $(HfO_2)_{1-x}(GeO_2)_x$ were highly similar, where the onset of crystallinity was slightly higher for HfO_2 films doped with SiO_2. However, the similarity in diffraction pattern does not translate into comparable dielectric behavior. We obtain a maximum dielectric constant of $\varepsilon_r = 28$ at 96 at% Hf for $(HfO_2)_{1-x}(SiO_2)_x$ whereas for $(HfO_2)_{1-x}(GeO_2)_x$ ε_r is only 19. The refractive indices vary linearly with composition for both systems. $(Ta_2O_5)_{1-x}(SnO_2)_x$ was crystalline only for the most Ta poor regions and remained amorphous for Ta cation fractions greater than 15%. Dielectric dispersion increased with increasing SnO_2 composition, which we assume is a result of defect and interstitial mediated conduction in SnO_2. Finally, anomalous features in the dielectric constant and refractive index were not evident in $(Nb_2O_5)_{1-x}(GeO_2)_x$. It is possible to obtain a high quality dielectric by doping low levels of GeO_2 into Nb_2O_5 and may be useful for applications requiring an amorphous oxide with a high dielectric constant.

4.2 INTRODUCTION

The wide application of thin film oxides is a testament to its importance in modern technology. They are used in optical coatings,[96] photocatalysis,[97] gate oxides,[26] and capacitors[2]. A relatively straightforward way to improve capacitor performance is to increase the dielectric constant of the insulator, ε_r, which can be achieved by stabilizing a high ε_r crystal structure by selective doping.[98, 99, 100, 36] Crystallization of the material is not necessarily deleterious, but in some applications we would like to prevent the materials from crystallizing. Thus, it would be desirable to enhance ε_r without inducing substantial crystallization. This effect has been observed for example, in hafnium and zirconium silicates with some controversy. [26, 37, 28] It may well be that the effects were real, but the disordered nature of these materials makes it difficult to understand the origin of these outstanding properties, so that the results can be replicated through tailored processing. Enhanced amorphous dielectric properties have also been observed experimentally in tantalum and niobium based oxides, such as $(Ta_{1-x}Nb_x)_2O_5$ and $(Ta_2O_5)_{1-x}(GeO_2)_x$ solid solutions.[101, 47] In the case of $(Ta_{1-x}Nb_x)_2O_5$, the dielectric constant jumped to $\varepsilon_r = 33$ at a Nb/(Nb+Ta) ratio of 42%, though the leakage current was markedly higher, and niobium oxide has a higher ε_r to begin with in comparison to tantala. [101] For $(Ta_2O_5)_{1-x}(GeO_2)_x$, the dielectric constant rose to $\varepsilon_r \sim 33$, strangely at a composition that is least GeO_2 rich.[47] Introduction of SiO_2 on the other hand does not induce such effect.[102]

The drastic differences in behavior even between materials doped with chemically similar oxides in itself is intriguing, and hence worth exploring. In this study, we examine the composition dependent dielectric and structural properties of $(HfO_2)_{1-x}(SiO_2)_x$, $(HfO_2)_{1-x}(GeO_2)_x$, $(Nb_2O_5)_{1-x}(GeO_2)_x$, and $(Ta_2O_5)_{1-x}(SnO_2)_x$ using off-axis reactive magnetron cosputtering. $(HfO_2)_{1-x}(SiO_2)_x$ is a system that has been studied heavily,

whereas $(HfO_2)_{1-x}(GeO_2)_x$ has garnered considerably less attention. We expect the two systems to behave similarly, owing to the chemical similarities between SiO_2 and GeO_2. Likewise, one would expect some similarity in the composition dependence of the dielectric properties of $(Nb_2O_5)_{1-x}(GeO_2)_x$, $(Ta_2O_5)_{1-x}(GeO_2)_x$, and $(Ta_2O_5)_{1-x}(SnO_2)_x$. Crystalline Nb_2O_5 tends to be a leaky oxide due to grain boundary conduction, which can be mitigated by forming an amorphous oxide to eliminate grain boundaries, or selective doping.[103, 101] The properties of $(Ta_2O_5)_{1-x}(SnO_2)_x$ have been characterized, but the composition resolution is low and the study was focused on Sn rich compositions, whereas the effect seen in $(Ta_2O_5)_{1-x}(GeO_2)_x$ occurs for GeO_2 rich compositions. [104]

4.3 EXPERIMENTAL METHODS

We fabricated the binary oxide compositions spreads in similar fashion to previously published experiments.[105] Films were sputtered on to 3 in. Si wafers precoated with e-beam evaporated 300 Å Cr and 350 Å Pt. Prior to film deposition the chamber was evacuated to a pressure of 10^{-6} Torr. The deposition condition for all films is summarized in Table 4.1.

Table 4.1: Deposition conditions for films sputtered in this experiment.

Oxide System	Power(W)	Gas	FlowRate(sccm)	Pressure
Hf-Si-O	100-20	Ar/O_2	15/10	30mTorr
Hf-Ge-O	100-20	Ar/O_2	15/10	30mTorr
Nb-Ge-O	100-20	Ar/O_2	15/10	30mTorr
Ta-Sn-O	100-30	Ar/O_2	15/10	30mTorr

The substrate was also intentionally biased at -75 V to ensure the formation of a dense film. Applying a substrate bias precludes the formation of columnar voids during deposition[61], which is undesirable when characterizing intrinsic film characteristics.

Substrate temperature during deposition was below 60°C.

A Bruker General Area Detector Diffraction System (GADDS) was used to collect X-ray diffraction intensities at several points on the wafer. Standard Cu-Kα x-rays (λ = 1.5418 Å) were used at 40kV and 40mA. The advantage of GADDS is the two-dimensional area detector which can captures a larger portion of the diffraction ring, otherwise not possible using a conventional diffractometer that is confined within a diffraction plane. We oriented the sample such that the captured image represents diffraction patterns corresponding to 2θ from roughly 20-70°. The detector position was fixed at 2θ = 40°, while ω = 20° and ϕ = 0°. Collection time was limited to 300 seconds achieving a high signal to noise ratio. The captured intensity was post-processed and integrated azimuthally in χ to generate a powder diffraction pattern.

We determined the film density by x-ray reflectivity (XRR) using a Rigaku Smartlab x-ray diffractometer as previously reported elsewhere. [102] Briefly, Cu Kα X-rays at 40mA/44kV were conditioned through x-ray optics. In this experiment we used a Ge-002 crystal monochrometer for improved accuracy. 1-D scans were conducted across the wafer at 21 equally spaced points along the long axis. At each point, scans were taken in the θ - 2θ geometry by varying the incident angle. Based on the x-ray beam footprint, we estimate the maximum error in composition for the density measurement to be roughly ±5%, though depending on the measured point the error was typically much smaller. Film density was directly evaluated by first determining critical angle θ_c [68] and using the following,

$$\rho = \frac{\theta_c^2 \pi A}{Z r_e \lambda^2 N_A} \tag{4.1}$$

where A is the atomic mass unit, r_e classical electron radius, λ the wavelength of the x-ray, N_A is Avogadros number, and Z the atomic number. We estimate the composition using previously determined deposition rates.

Index of refraction and film thickness were determined using a J.A. Woollam Variable Angle Spectroscopic Ellipsometer. The ellipsometric parameters ψ and δ were measured at 65, 70, and 75°. At each angle the wavelength was varied from 450 - 1000 nm at 100 nm increments. A film/substrate bilayer model was used to fit ψ and δ. The optical constants of the substrate layer was determined prior to film deposition and was fixed during the fitting process. The film was modeled using a two-term Cauchy relation, $n(\lambda) = A + \frac{B}{\lambda^2}$. The WVASE32 software package supplied by J.A. Woollam was used to fit A, B, and thickness by recursively varying the parameters to minimize the mean squared error. The refractive index was taken to be A, which is the refractive index at the infinite wavelength limit well away from absorption edge.

Electrical characterization was performed by preparing a metal-insulator-metal structure (MIM). Top electrodes (200 micron diameter, 350 Å thick) were electron beam evaporated on to the film through a shadow mask. Contact to the top electrode was made using a BeCu tip while gold wire was soldered to the bottom electrode using indium solder. Capacitance and $\tan\delta$, i.e. $\frac{Re(\epsilon)}{Im(\epsilon)}$, was measured across the MIM capacitor using a HP 4284 precision inductance-capacitance-resistance (LCR) meter at 100 mV over the frequency range of 100 Hz - 100 kHz.

4.4 RESULTS AND DISCUSSION

4.4.1 $(HfO_2)_{1-x}(SiO_2)_x$ and $(HfO_2)_{1-x}(GeO_2)_x$

We first present the results for the diffraction measurements. The strong reflections at roughly 2θ = 39.5°, 46°, and 67°observed in all samples correspond to the (111), (200), and (220) diffraction planes respectively generated by the Pt bottom electrode.

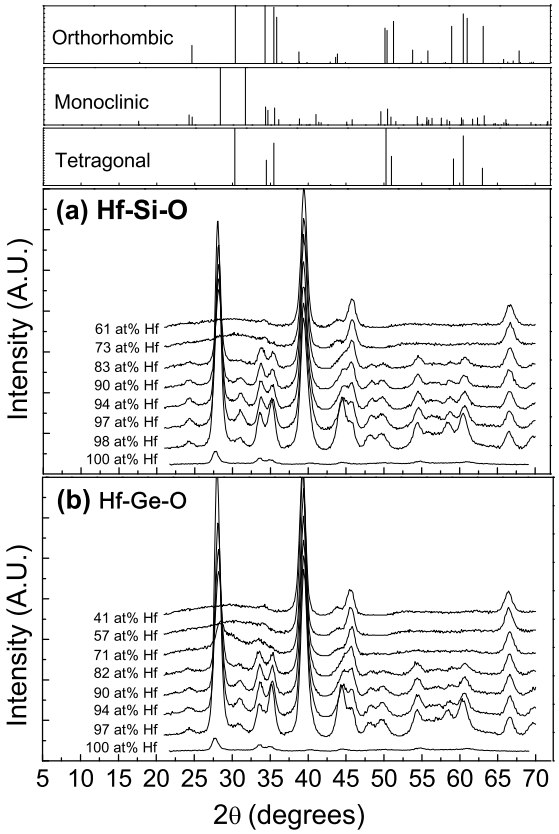


Figure 4.1: The XRD spectra for (a) $(HfO_2)_{1-x}(SiO_2)_x$ and (b) $(HfO_2)_{1-x}(GeO_2)_x$ plotted for several different Hf cation compositions. The observed patterns between Si and Ge doped HfO_2 are highly similar, demonstrating their chemical resemblance. The diffraction pattern for the mixed films show that the monoclinic phase is the dominant phase, however comparison with available diffraction patterns suggest the presence of other metastable phases. For reference the diffraction pattern of pure HfO_2 is shown at the bottom. The weak reflections reveal the presence of partially crystalline domains.

Additionally, the recurring peaks at 44° corresponds to the Cr (110) underlayer, while the peak at 34° is a detector artifact.

For the hafnium based systems a change in composition elicits a clear shift in the structure from amorphous (HfO_2 poor compositions) to polycrystalline (HfO_2 rich compositions), as seen in Figure 4.1. As a reference we also present the diffraction pattern of pure HfO_2 deposited under similar conditions. The pattern displays very weak reflections signifying the presence of small crystalline domains embedded in a mostly disordered matrix.

It is not unusual that the doped hafnia films deposited at low temperatures show sign of crystallinity. Thicker HfO_2 films may crystallize [106] (our films are 100 - 300 nm thick) and deposition conditions [107, 108] play a role in whether HfO_2 becomes amorphous or crystalline. Even for films that seem to be amorphous HfO_2 thin films, careful inspection of its structure reveals that it exhibits a certain level of medium-range order, a characteristic of a fairly poor glass former.[109] Crystallinity decreases with increasing substitution of glass formers SiO_2 and GeO_2 and is consistent with our intuition that substitution of glass formers should increase the crystallization temperature of the film.[110]

A direct comparison of the diffraction patterns between the two systems shows an uncanny resemblance, demonstrating the chemical similarities between SiO_2 and GeO_2. Continuous rings are observed in the area detector pattern, which indicate that the grains are equiaxed with no preferential orientation. Comparison with powder diffraction and calculated diffraction patterns retrieved from the Inorganic Crystal Structure Database (ICSD) with the integrated powder diffraction patterns show that the HfO_2 rich crystalline films primarily consist of the monoclinic phase, indicated definitively by the peak at 28.2°. Note, the bulk equilibrium phase of HfO_2 at room temperature is monoclinic.

However, we see the peaks are not well aligned with a pure monoclinic phase. The structural deviation is reflected in the dielectric behavior of the two systems as a function of composition (Figure 4.2). For $(HfO_2)_{1-x}(SiO_2)_x$, we see that a rough extrapolation of ε_r to pure SiO_2 in the amorphous region ($65 <$ at% Hf < 75) points to about 3.9. A linear extrapolation in the other direction to pure HfO_2 (denoted by the dashed line) yields ε_r of 22, the dielectric constant of pure amorphous HfO_2.[111] At the onset of crystallization in the region $75 <$ at% Hf < 85, we observe a slope change in ε_r (dotted line) such that now ε_r extrapolates to the dielectric constant of monoclinic HfO_2, $\varepsilon_r = 16$.[93]

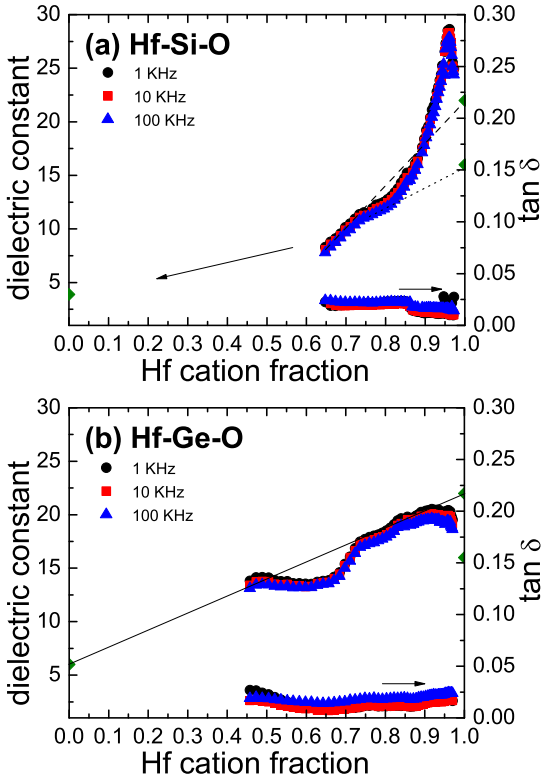


Figure 4.2: The dielectric constant and dissipation factor of (a) $(HfO_2)_{1-x}(SiO_2)_x$ and (b) $(HfO_2)_{1-x}(GeO_2)_x$ as function of Hf cation composition. In contrast to the XRD patterns in Figure 4.1, the dielectric behavior of the two materials are different.

Further addition of HfO_2 causes another slope change at ~85 at% Hf accompanied by a discontinuity in tan δ. In this region ε_r is enhanced dramatically to a maximum dielectric constant of approximately 28 at a concentration of 96 at% Hf. The dielectric behavior in this region is clear indication that the monoclinic phase is not the sole phase within this composition range, since ε_r of a pure monoclinic phase is only 16. Finally, the dielectric constant drops down for even higher Hf concentrations.

The dielectric behavior of $(HfO_2)_{1-x}(GeO_2)_x$ is clearly different from $(HfO_2)_{1-x}(SiO_2)_x$, even though their diffraction patterns are highly similar. At 50 at% Hf, the dielectric constant is consistent with the linear interpolation (solid line), between the endmember values for pure amorphous HfO_2 and GeO_2. Addition of Hf does not increase the dielectric constant and consequently, ε_r is suppressed below the linear interpolation for

50 < at% Hf < 70. At the onset of crystallization (~70 at% Hf) the dielectric constant once again becomes collinear to the linear interpolation. Increasing the Hf concentration further does not result in any kind of enhancement; the dielectric constant of the Hf richest composition is 19, roughly between the amorphous and monoclinic dielectric constant.

Without further analysis, it is difficult to determine the cause for the trends seen in Figure 4.2. Nevertheless, the enhancement of ε_r observed in $(HfO_2)_{1-x}(SiO_2)_x$ is unfeasible without a stabilization of high ε_r phases such as the tetragonal, or cubic like phase. It has been shown that both Si and Ge are effective stabilizers of the tetragonal phase, where Si is slightly more effective than Ge, i.e. lower concentrations of the dopant is required.[112, 36] Its effectiveness in part may be attributed to ionic radii where smaller dopants such as Si and Ge tend to stabilize the tetragonal phase.[113] For stabilization of the tetragonal phase, calculations by Lee et al.[113] predict a minimum concentration of 10.9% for Si and 18.6 % for Ge. Furthermore, when considering temperature as a parameter, Debernardi predicts that at 320 K, 12.5% Ge is required to stabilize the tetragonal phase.[114] Our data suggests that it is fairly easy to stabilize a tetragonal-like phase when Si is the dopant, while Ge seems less effective and may require additional processing. Deviations from predicted values may be caused by stress and grain size effects, which shift the transformation temperature. [115] It is possible that the deposition process and the dopant in combination have created stresses in the film causing the variations.

We also present the density and the refractive indices in Figures 4.3 and 4.4 respectively as a function of Hf composition. The density for both systems increases linearly with increasing Hf composition in the amorphous regions. Crystallization of the film is marked by a clear jump in the density. The onset of crystallinity for $(HfO_2)_{1-x}(GeO_2)_x$

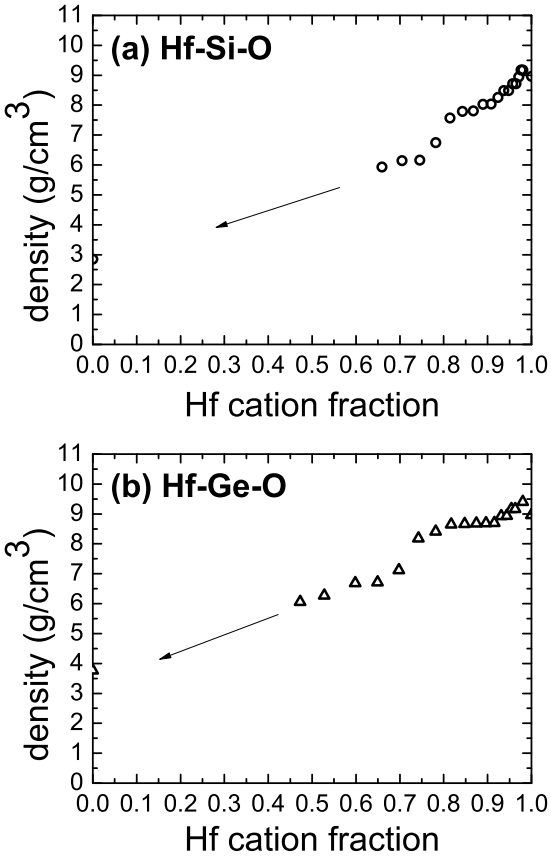


Figure 4.3: The density of (a) $(HfO_2)_{1-x}(SiO_2)_x$ and (b) $(HfO_2)_{1-x}(GeO_2)_x$ as function of Hf cation composition determined using XRR. Crystallization of the film causes the film density to jump dramatically.

is slightly lower in composition in comparison to the $(HfO_2)_{1-x}(SiO_2)_x$; approximately 70 at% Hf for Ge doped HfO_2 versus 80 at% Hf for Si doping. This suggests the slightly higher effectiveness of SiO_2 as a glass former compared to GeO_2. We find that although there is a general correlation between increased density and dielectric constant, there is no direct evidence which explains the large disparity in dielectric behavior between $(HfO_2)_{1-x}(SiO_2)_x$ and $(HfO_2)_{1-x}(GeO_2)_x$. For example, direct comparison of the density of both systems at around 96 at% Hf, the density of $(HfO_2)_{1-x}(SiO_2)_x$ is 8.72 g/cm^3, whereas the density of $(HfO_2)_{1-x}(GeO_2)_x$ is 9.17 g/cm^3. This in turn implies that substantial differences in the phononic contributions may be at play.

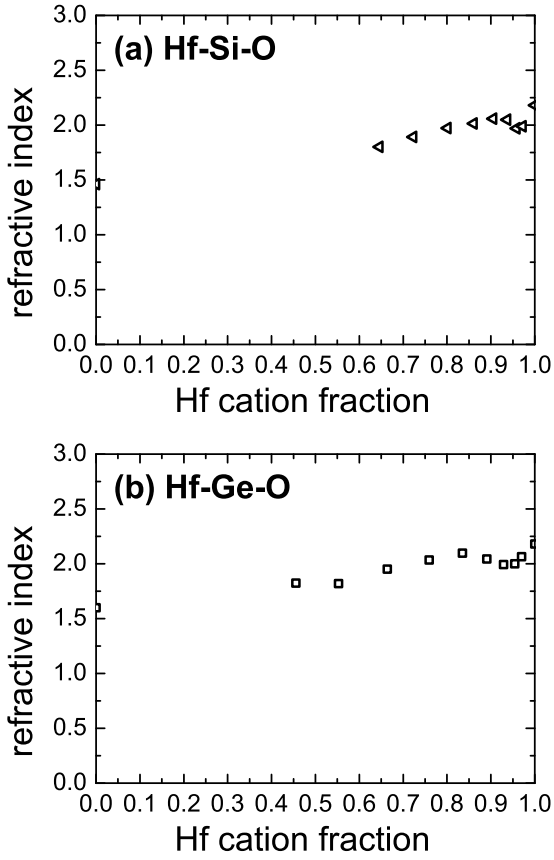


Figure 4.4: Refractive index of (a) $(HfO_2)_{1-x}(SiO_2)_x$ and (b) $(HfO_2)_{1-x}(GeO_2)_x$ as function of Hf cation composition.

The refractive index varies linearly with Hf composition for both systems. For the Hf richest compositions a slight depression from a linear trend is observed. A large variation in the refractive index for HfO_2 is not uncommon,[107] but a more likely contribution is the increase in roughness with crystallinity,[106] indicated by a larger mean squared error (MSE) in the fitting, as our simple model does not take into account surface roughness.

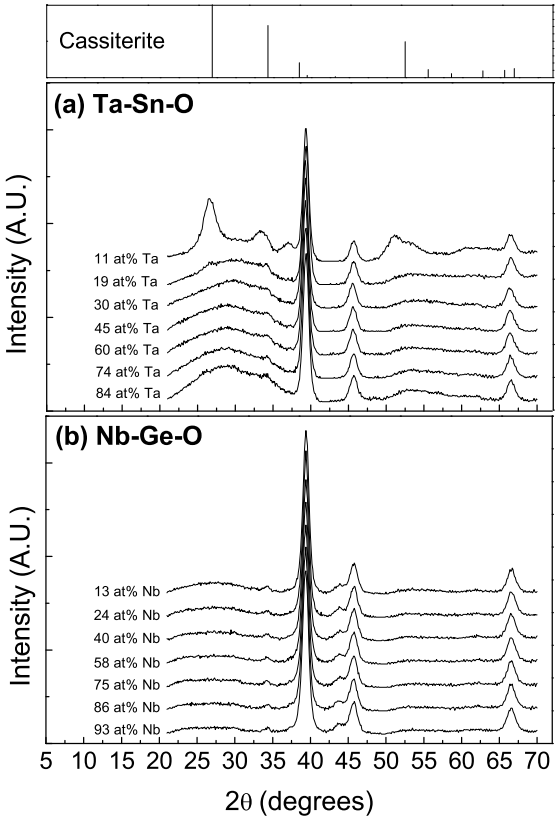


Figure 4.5: XRD spectra for (a) $(Nb_2O_5)_{1-x}(GeO_2)_x$ and (b) $(Ta_2O_5)_{1-x}(SnO_2)_x$ plotted for several cation compositions. The lack of any discernible diffraction peaks indicate that the $(Nb_2O_5)_{1-x}(GeO_2)_x$ is x-ray amorphous for the entire composition range probed. $(Ta_2O_5)_{1-x}(SnO_2)_x$ is x-ray amorphous for high Ta_2O_5 rich composition, but crystallizes at high SnO_2 concentrations.

4.4.2 $(Nb_2O_5)_{1-x}(GeO_2)_x$ and $(Ta_2O_5)_{1-x}(SnO_2)_x$

The XRD patterns for $(Nb_2O_5)_{1-x}(GeO_2)_x$ and $(Ta_2O_5)_{1-x}(SnO_2)_x$ are shown in Figure 4.5. Patterns for $(Nb_2O_5)_{1-x}(GeO_2)_x$ show that it is amorphous for all compositions investigated, whereas crystallization is observed for tantalum poor compositions in $(Ta_2O_5)_{1-x}(SnO_2)_x$. Comparison with Power Diffraction File (PDF) suggest that the peaks embody the symmetry of the tetragonal (cassiterite) phase.

The sensitivity of the dopant concentration on the dielectric behavior of $(Ta_2O_5)_{1-x}(SnO_2)_x$ is shown in Figure 4.6, where two distinct regions of dielectric be-

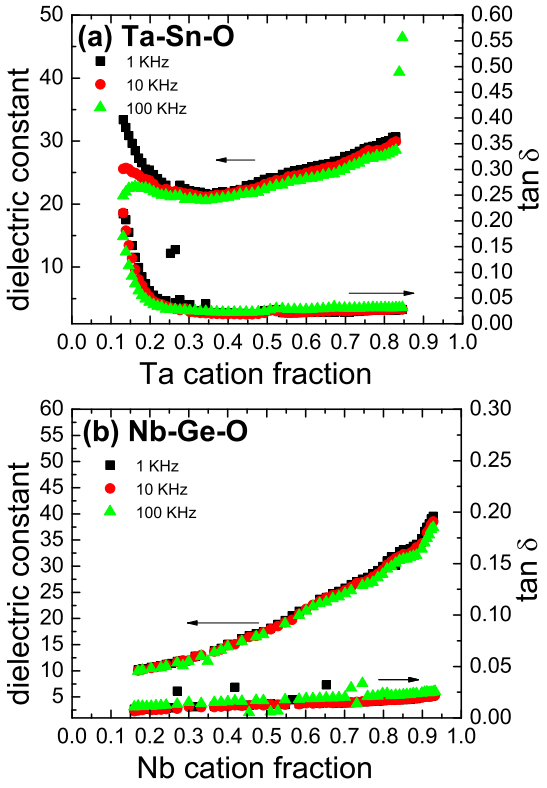


Figure 4.6: The dielectric constant and dissipation factor for (a) $(Nb_2O_5)_{1-x}(GeO_2)_x$ and (b) $(Ta_2O_5)_{1-x}(SnO_2)_x$ plotted as a function of composition. The dielectric constant of $(Nb_2O_5)_{1-x}(GeO_2)_x$ shows a mostly linear dependence in ε_r with composition, whereas $(Ta_2O_5)_{1-x}(SnO_2)_x$ shows a nondispersive and highly dispersive region.

havior can be distinguished. For high Ta_2O_5 concentrations we observe a near linear dependence on the dielectric constant and in this region, the dissipation factor is also low ($\tan\delta < 0.1$) indicating a high quality insulating film. Substitution of an aliovalent cation does not seem to promote a defect mediated transport mechanism such as hopping conductivity, as suggested by the low dissipation factor. Similarly, the substitution of ZrO_2 in Ta_2O_5 decreases the conductivity of Ta_2O_5 phase, the opposite of what is expected for a classical point defect model.[116] The dielectric constant of the most Ta_2O_5 rich composition is around 27 at approximately 85 at% Ta. Linearly extrapolating to pure Ta yields a dielectric constant of approximately $\varepsilon_r = 31$ which is enhanced over that of pure amorphous Ta_2O_5 ($\varepsilon_r = 22$),[61] while extrapolation to pure

SnO_2 gives $\varepsilon_r = 13$, consistent with literature values.[117] With further incorporation of SnO_2 ε_r becomes dispersive, evidenced by a high dissipation factor ($\tan\delta > 0.1$) and the highly frequency dependent nature of ε_r. This is not surprising since SnO_2 is a conductive oxide.[118] Diffraction patterns (Figure 4.5) show that the onset of crystallization correlates with the dispersive behavior, hence the dispersive nature of the film is likely caused by conduction within the SnO_2 rather than defects in Ta_2O_5.

The dielectric constant of amorphous $(Nb_2O_5)_{1-x}(GeO_2)_x$ scales linearly with composition, with low dissipation factors, and shows no sign of dispersive behavior (Figure 4.6). At an Nb cation concentration of 93% $\varepsilon_r \sim 42.5$, while extrapolation of ε_r to pure Nb_2O_5 indicates a ε_r for pure $Nb_2O_5 \sim 47 - 48$, which is comparable with ε_r of fully oxidized Nb_2O_5 ($\varepsilon_r = 45 - 50$).[101, 103] Though the dielectric constant may not be dramatically enhanced over the end member, an insulating amorphous Nb_2O_5 thin film oxide may still be of technological interest, in particular since it is a conductive oxide,[119] that is sensitive to oxygen partial pressure during deposition.[103] Because conductivity in Nb_2O_5 is inversely related to oxygen partial pressure it is often attributed to oxygen ion vacancies in Nb_2O_5 creating charge carriers that are thermally excited into the conduction band.[119] Doping aliovalent cations often create sites that enable dielectric relaxation via hopping. For instance, previous measurements on sintered $(Nb_2O_5)_{0.92}(SiO_2)_{0.08}$ bulk ceramic samples exhibited significant frequency dependent dielectric relaxation at room temperature. [120] Three distinct mechanisms were identified and the authors hypothesized that the relaxation species composed of hopping ions between the Nb and the substituted Si ion and the large dispersion is accompanied by a substantially higher dielectric constant ($\varepsilon_r \approx 125$ at room temperature).[120]

Even with the introduction of GeO_2 into Nb_2O_5, the thin films exhibit low dissipation factors indicating the absence of a deleterious relaxation mechanism. Nonethe-

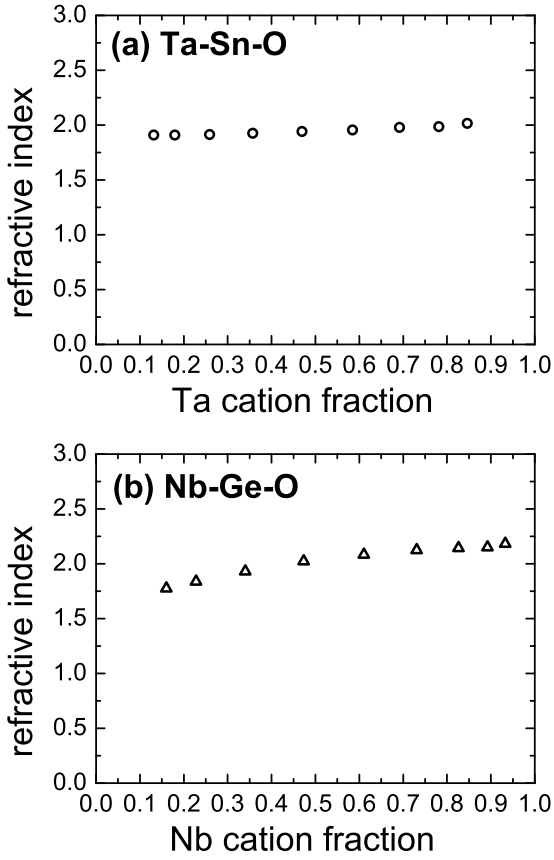


Figure 4.7: Refractive index of (a) $(Nb_2O_5)_{1-x}(GeO_2)_x$ and (b) $(Ta_2O_5)_{1-x}(SnO_2)_x$ as function of Hf cation composition.

less, direct comparisons are difficult to make since the $(Nb_2O_5)_{0.92}(SiO_2)_{0.08}$ was a bulk crystalline ceramic, whereas our materials were amorphous thin films. Bulk crystalline ceramics contain grains and grain boundaries, which can contribute to several types of conduction mechanisms, which are absent in amorphous materials.

Anomalous behaviors were not observed for the refractive indices for both systems (Figure 4.7) and the refractive index scales with composition. This trend is also reflected in the densities calculated from x-ray reflectivity (Figure 4.8). The density of the endmember values were also calculated directly from pure films deposited on to Si. Crystallization of the $(Ta_2O_5)_{1-x}(SnO_2)_x$ film at 0.14 Ta cation fraction causes the film density to jump to 6.78 g/cm^3 and is consistent with diffraction patterns and dielectric

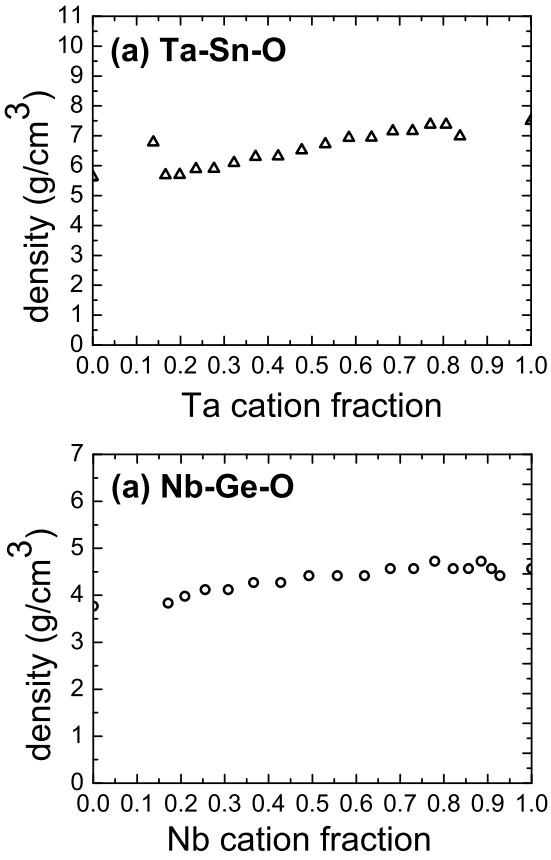


Figure 4.8: The density of (a) $(Nb_2O_5)_{1-x}(GeO_2)_x$ and (b) $(Ta_2O_5)_{1-x}(SnO_2)_x$ as function of Hf cation composition determined using XRR. The linear variation in density for $(Nb_2O_5)_{1-x}(GeO_2)_x$ as a function of composition reflects the amorphous nature of for all composition. $(Ta_2O_5)_{1-x}(SnO_2)_x$ densities also vary linearly except for the point where the film crystallizes.

measurements. This makes it appear that the density is nonlinear, however the density of pure amorphous SnO_2 as seen in Figure 4.8 is 5.62 g/cm^3 and is consistent with the linear trend. Since the refractive index reflects the electron density of the material increase in density should cause the refractive index to rise. Our refractive index measurements for $(Ta_2O_5)_{1-x}(SnO_2)_x$ do not go low enough in Ta cation fraction and thus we were not able to observe an increase in the refractive index.

4.5 Conclusion

Using off-axis rf sputtering, we have studied the physical and dielectric properties of four pseudo-ternary oxides of varying composition: $(HfO_2)_{1-x}(SiO_2)_x$, $(HfO_2)_{1-x}(GeO_2)_x$, $(Nb_2O_5)_{1-x}(GeO_2)_x$, and $(Ta_2O_5)_{1-x}(SnO_2)_x$. Both $(HfO_2)_{1-x}(SiO_2)_x$ and $(HfO_2)_{1-x}(GeO_2)_x$ crystallized with increasing HfO_2 compositions and became amorphous with increasing concentrations of $Si(Ge)O_2$. The diffraction patterns of the two systems were similar, but the dielectric constants varied differently with composition. A clear peak in the dielectric constant can be observed in the $(HfO_2)_{1-x}(SiO_2)_x$ films, with a maximum dielectric constant of $\varepsilon_r = 28$ at 96 at% Hf, while the dielectric constant of $(HfO_2)_{1-x}(GeO_2)_x$ plateaus around $\varepsilon_r = 19$. Reflectivity measurements were consistent with the diffraction measurements and provided further detail into the amorphous-crystalline phase transition. The onset of crystallinity is slightly higher for the SiO_2 doped films; 80 at% Hf versus 70 at% for the GeO_2 doped films. In contrast to the dielectric constants, the refractive indices varies roughly linearly with composition.

$(Ta_2O_5)_{1-x}(SnO_2)_x$ was amorphous for SnO_2 poor compositions and crystallized into the cassiterite phase only for the most SnO_2 rich composition evaluated, and is indicated by a sudden increase in density. Increased dielectric dispersion and dissipation factor with increasing SnO_2 suggest that the film is becomes more conductive, where conduction is most likely through polycrystalline SnO_2 grains. Finally, $(Nb_2O_5)_{1-x}(GeO_2)_x$ was amorphous for all measured compositions and showed no anomalous features in the dielectric constant and refractive index. Not surprisingly, the density of the solid-solution changes linearly with composition. Doping low levels of GeO_2 into Nb_2O_5 resulted in a high quality dielectric and may be useful for applications requiring a high dielectric constant oxides.

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CHAPTER 5

SUMMARY AND FUTURE OUTLOOK

In this dissertation, we have evaluated the properties of various complex amorphous thin film oxides synthesized by combinatorial sputtering. We mostly focused on two aspects of materials science: structure and property. Structurally, amorphous materials are inherently hard to characterize and throughout this study, we learned that this is especially true for thin film amorphous materials. However, by combining several different techniques we were able to effectively characterize their structure at both the macroscopic and local level. Taking what we learned structurally and what we learned by probing their dielectric properties, we learned something about the structure-property relations of the complex amorphous oxides.

In chapter 3, we found that the structural and dielectric properties of two chemically similar amorphous oxides, namely Ta-Ge-O and Ta-Si-O, were significantly different. We saw that the dielectric constant of Ta-Ge-O was dramatically enhanced over linear interpolation of its end members, while Ta-Si-O did not exhibit this behavior. This type of enhancement was not observed in the refractive index for both materials. We used x-ray reflectivity to directly probe the density of the thin films as a function of composition. This technique allowed us to calculate the polarizabilities of the oxides without relying on tabulated values, avoiding any additive assumptions of the individual polarizabilities. We also explored the limitations of the Clausius-Mossotti relation applied to the two systems and found that the use of tabulated atomic polarizabilities and experimentally inferred volumes systematically underestimates the dielectric constant in both systems. This result highlights the importance of directly calculating the densities and atomic polarizabilities. We then used FTIR to explore the changes in active phonon modes of Ta-Ge-O as a function of composition. The spectra did not show any obvious

features in the frequency range that would explain the anomalous dielectric behavior of Ta-Ge-O. It is likely that the phonon modes contributing to the enhancement exist at lower wavenumbers. Therefore, it might be worthwhile to conduct a careful far-infrared spectroscopy experiment to cover the sub 400 cm^{-1} range in future studies. A Raman spectroscopy study may also be a good complement to FTIR.

In chapter 4, we extended the study in chapter 3 by probing the atomic structure of pure Ta_2O_5 and Ta-Ge-O using high energy synchrotron x-ray diffraction and absorption methods. Simultaneously applying different techniques greatly reduced the uncertainty in structure space by constraining the possible structural parameters of the local environment. To reduce scattering from the substrate, we sputtered the thin films on silicon nitride windows. We found bond lengths and coordination numbers of amorphous Ta_2O_5 to be similar to values reported in previous studies, where the Ta-O coordination was found to be lower than those measured in crystalline Ta_2O_5 compounds. Past research suggests this is due to 1) Ta-O having a fairly low field strength (for example compared Si-O which is a good network former) 2) the Ta cation is large in size.

Doping of GeO_2 in Ta_2O_5 result in substantial changes to the local environment. We identify two distinct regions in composition based on pair bond distance and coordination data. The Ta-O/Ge-O bond distance does not differ greatly from pure Ta_2O_5 for high Ta cation fractions; only when a considerable amount of GeO_2 is added does $r_{Ta-O/Ge-O}$ begin to decrease to r_{Ge-O}. Our EXAFS data also shows that the Ta-O structural unit experiences an anisotropic distortion due to Ge doping. We find that the most dramatic change in the local environment around Ta occurs for the metal-metal (M-M) pairs, namely Ta-Ta and Ta-Ge. Further, the trend in the O-O pair bond distance was explained as a consequence of a high Ta-Ge coordination at fairly low GeO_2 composition. When the composition of Ta and Ge cations are roughly equal, the short-range

order starts to resemble that of low temperature crystalline Ta_2O_5, evidenced by the coordination number of the Ta-O inner shell. Moreover, we surmise that a major factor in the reduction of density observed in chapter 3 is due to the change in coordination of inner M-M pairs, while there is only minimal contribution from changes in M-O and M-M bond distances. This suggests that network conformation has a larger effect on the density than simple changes in bond length. **need to write a little more here**

In chapter 5, we studied the composition dependence of the dielectric properties of another complementary system, Zr-Al-O and Zr-Si-O. Specifically, we used these two systems to systematically investigate the effect of the presence or absence of crystallinity on the dielectric constant and breakdown field. An essentially linear variation of the static dielectric constant, ε_s, was observed as a function of composition, x, for compositions rich in the p-block element, i.e., $x > 0.4$, for both chemical systems. However an abrupt change in ε_s is found near $x \approx 0.35$, associated with the onset of crystallinity in as-deposited films. Measurements of the index of refraction at optical frequencies, combined with a simple Clausius-Mossotti interpretation, indicates that low-frequency (ionic) contributions to the polarizability exhibit systematic deviation with respect to values linearly interpolated from the endmembers. These trends are not consistently affected by the presence of crystalline order, but are related to changes associated with heterogeneous local oxygen coordination and bonding. There were no significant differences in the breakdown field associated with the level of crystallinity in both systems.

In chapter 6, we expand on the study in chapter 3 by substituting different cations in the Ta/Ge cation site which include the combinations Hf-Si-O, Hf-Ge-O, Ta-Sn-O, and Nb-Ge-O. Hf-Si-O and Hf-Ge-O showed similar trends in the x-ray diffraction spectra as a function of composition. Both remain amorphous until a critical Hf cation concentration is reached, upon which crystallizing into a mostly monoclinic phase mixed with

possibly tetragonal and orthorhombic phases. Crystallization of the oxides were clearly reflected in the density measurements as jumps in the density for both systems. However, when we probed the dielectric constant of each system as a function composition, the two systems show dissimilar trends in ε_r, namely Hf-Si-O shows an enhanced ε_r of approximately 28 at a concentration of 96 at% Hf, while ε_r for Hf-Ge-O is not enhanced for any composition. The dielectric enhancement observed for Hf-Si-O is clear indication that the monoclinic phase is not the sole phase within this composition range, since ε_r of a pure monoclinic phase is only 16. The refractive index is linear with composition for both Hf-Si-O and Hf-Ge-O.

Ta-Sn-O and Nb-Ge-O are quite different structurally and electronically compared to Hf-Si-O and Hf-Ge-O. Inspection of the diffraction spectra indicates that both systems remain amorphous for most of the compositions probed in the study. The exception is for the Sn rich region in Ta-Sn-O (< 30 at% Ta), where SnO_2 crystallizes into a tetragonal (cassiterite) phase. Crystallization of SnO_2 is clearly reflected in the dielectric constant below 30 at% Ta by two measures: the increased dispersion in the dielectric factor and the marked increase in dissipation factor. For compositions > 30 at% Ta, Ta-Sn-O remains amorphous for all compositions investigated ($\sim$ 15 at% Ta < x < $\sim$85 at% Ta). Nb-Ge-O is x-ray amorphous for $\sim$15 at% Ta < x < $\sim$ 95 at% Ta, thus not surprisingly, the dielectric constant and the refractive index scales linearly with composition. It is interesting to note that the dielectric constant in the Nb rich side does not become dispersive since pure Nb_2O_5 films are generally conducting. For a more detailed analysis of the dielectric behavior of the oxide, it may be worthwhile to investigate the relaxation mechanism of the oxide as a function of frequency. The density of Ta-Sn-O and Nb-Ge-O varies linearly with composition with the exception of the jump in density for Ta-Sn-O, where SnO_2 crystallizes.

There are several different paths we can go down from here. An obvious choice would be to probe the local environment around Ge in Ta-Ge-O using EXAFS. Another low hanging fruit would be to use high energy x-ray diffraction and absorption methods to study Ta-Si-O. Stuctural measurements can be complimented with infrared spectroscopy to directly observe the changes in phonon modes as a function of Ta cation composition for Ta-Ge-O and Ta-Si-O, to establish a more rigorous structure-property relationship. Obviously we are not limited to Ta based systems; we can apply our composition spread approach to rapidly analyze the local structure variation of several different amorphous systems, including ternary oxides. Finally, we can utilize these materials in novel device applications to study changes in device performance with composition. For example, it would be interesting to study the effects of composition in resisitive memory and logic devices such as memristors.

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